

SINGAPORE POLYTECHNIC

**SCHOOL OF ARCHITECTURE AND THE BUILT
ENVIRONMENT**

Diploma in Facilities Management

Streamlining Defects Reporting Process

Yip Li Tong	2225546
-------------	---------

Krystal Tham Kai Lin	2225559
----------------------	---------

Nur Aliyah Umairaa Binte Riduan	2225757
---------------------------------	---------

Year of Study: Year 3

Project Facilitator: Mr Ubaidullah

24 January 2025

Project Summary

2024 S2 DFM Integrated Project

DFM-IP Summary

A. Project Title:

Streamlining Defects Reporting Process

B. Project Students:

	Admin No.	Class	Name	Remarks
1.	2225546	DFM/FT/2B/02	Yip Li Tong	Leader
2.	2225559	DFM/FT/2B/02	Krystal Tham Kai Lin	Secretary
3.	2225757	DFM/FT/2B/02	Nur Aliyah Umairaa Binte Riduan	Time & Action Keeper

C. Project Facilitator:

Mr. Ubaidullah

D. Collaborated with: (if any, insert organisation's details)

Collaboration with Property Facility Services Private Limited

E. Project Abstract:

In not more than 150 words, summarise your project with objectives, scope of work, deliverables and conclusion.

The purpose of this app is to streamline the defect reporting process, resolving inefficiencies and disorganisation observed during internship, in which WhatsApp messages were used to monitor defects. The goal is to improve Facilities Management (FM) operations by adding additional features from the existing software in the market.

The group have created an application to program and integrate the original concept. This include designing the graphical user interface to be user-friendly for all ages. The application consists of form, dashboards, tracking of asset and inventory, checklist and calendar.

We use a software (Power App) to create an application. Due to the lack of coding knowledge, we referred to sources like YouTube and electronics books on how to code.

Overall, this application may facilitate smooth FM operations, making it easy to utilise for all ages.

F. Project Photos or Figures: Insert 6 photos or figures here to illustrate the salient points of your project best.

The image displays two screenshots of the GLIDIFY mobile application interface.

Left Screenshot: Defect Reporting Form

- Name:** YIP LI TONG
- Reporting Department:** Which department are you from?
- Building Name:** Which building is the defect found?
- Level:** Which storey?
- Area:** Specific area of the level?
- Priority:** Critical
- Due Date:** 10/01/2025 03:46:24

Right Screenshot: Task View

- Task 1:** Critical, Electrical & Lighting, Completed, 09/01/2025 02:37:00
- Task 2:** Critical, Electrical & Lighting, In Progress, 09/01/2025 02:32:16
- Task 3:** Critical, Scratches / Damages, Cannot fix, 09/01/2025 01:10:14
- Task 4:** Critical, Electrical & Lighting, Pending, 07/01/2025 17:10:24
- Task 5:** Critical, Air-conditioning, Completed, 06/01/2025 15:04:59
- Task 6:** Minor

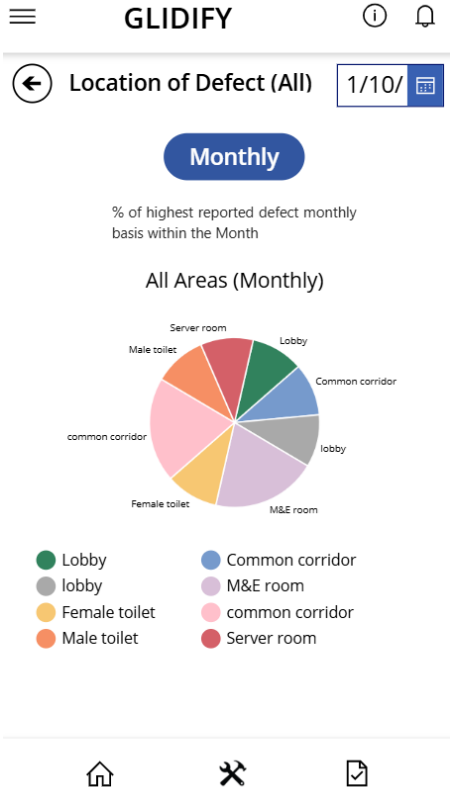
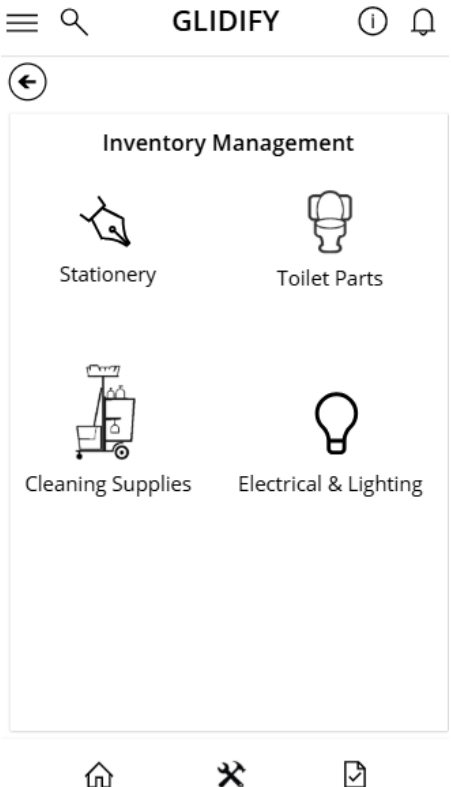
Figure 1: Reporting Defect Form User's Name and due date are default. The due date set based on the priority.	Figure 2: Task View Colour-coded label (based on task expiry and working progress) to enhance visuality
	
Figure 3: Dashboard Analyses and highlights summary of data (help manager to identify most occurred defect by date)	Figure 4: Assets Inventory keep track of the stock listing and ordering list by allowing user updating and storing of data in backend database



Figure 5: Daily Checklist

A day-to-day checklist for users to use to tick off whenever they perform their routines

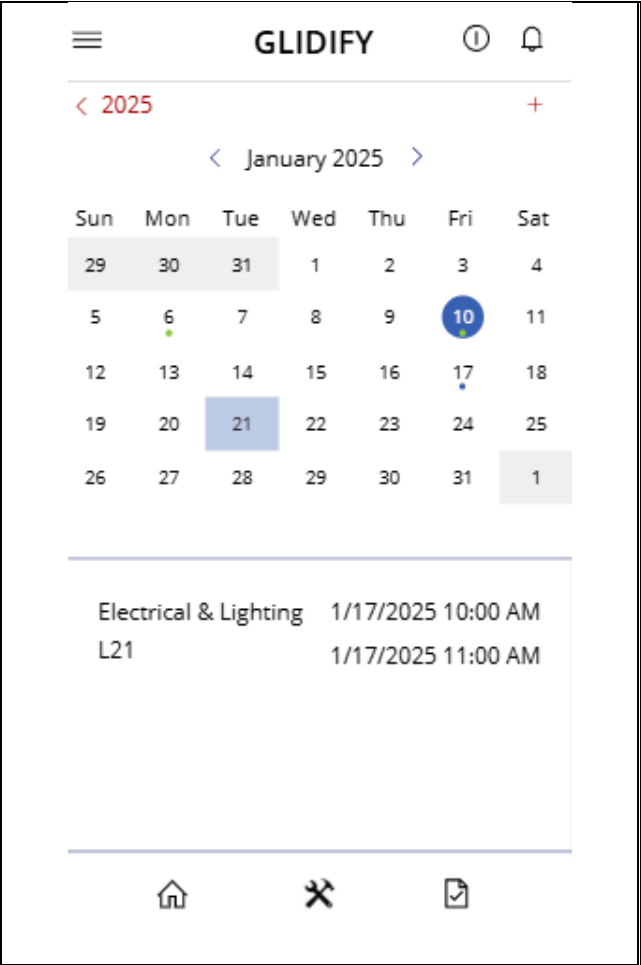


Figure 6: Scheduled Maintenance

Able to review details base on date selected by clicking on the numbers above

Acknowledgement

Throughout the process of planning, researching, designing and finalising our application, we have received supports and advice from our peers and facilitators. We are truly grateful for their guidance and encouragement throughout this journey.

We would like to express our sincere gratitude to our facilitator, Mr Ubai, for this encouragement. The constant feedback and support he has given us have improved our skills and broaden our knowledge related to Facilities Management (FM). We are honourable to have the opportunity to work with an exceptional knowledgeable and well-connected lecturer.

Besides that, we would also like to thank Mr Low Check Choon and Mr. Lim Boon Tech from the IT support for their assistance in troubleshooting and resolving bugs in our application. They have provided us with in-depth programming knowledge which was essential in refining and enhancing the quality of our application.

Last but not least, we would like to express our gratitude to our Industrial Company, Mr. Muhammad Fariz from Property Facility Services Pte Ltd in supporting our Final Year Project. This partnership has provided us a constructive review from a different perspective which added invaluable meanings.

Table of Contents

Project Summary	ii
Acknowledgement	vi
List of Figures	ix
List of Tables	x
Chapter 1 – Introduction	1
1.1 Context of Problem.....	1
1.1.1 Initial Problem.....	1
1.1.2 Issue Arise by Companies	1
1.2 Objectives of Application.....	1
1.2.1 Initial Objectives	1
1.2.2 Final Objectives.....	2
1.3 User of the Application	2
1.3.1 Primary User	2
1.3.2 Secondary User.....	2
1.4 Proposal for Industrial Linkage	3
1.5 Team Planning.....	3
Chapter 2 – Research.....	5
2.1 22-weeks internship	5
2.1.1 Origin and Development of Idea	5
2.1.2 Implementation of Solution.....	5
2.2 Optimising Solution.....	6
2.3 Considerations for Application Development	7
2.3.1 Key Considerations in Selecting an App-Building Platform	7
2.3.2 Research on App building platforms.....	8
2.3.3 Refining Choices	13
2.3.4 Comparison between AppSheet and PowerApps.....	13
2.4 Additional Research	15
2.4.1 Research on Application Features	16
2.4.2 Research on Defect Form Questions.....	16
2.4.3 Research on Coding.....	16
2.4.4 Research on Graphical User Interface (GUI)	17
Chapter 3 – Features.....	19
3.1 Phase 1.....	19
3.1.1 Defect Reporting Form	19
3.1.2 Task View.....	22
3.1.3 Task Information.....	27
3.1.4 Dashboard.....	30
3.2 Phase 2.....	33
3.2.1 Asset & Inventory	33
3.2.2 Daily checklist	43

3.2.3 Scheduled Maintenance	47
3.3 Fixed Features	55
3.3.1 Interface Panel	55
3.3.2 Notifications.....	57
3.3.3 Profile Page	60
3.3.4 Help Information	63
3.3.5 Home Page	65
3.3.6 AI Chatbot (FAQ)	67
3.3.7 Database	68
Chapter 4 – Problem & Skills Gap	70
4.1 Skills Gap	70
4.2 Unavailability of extended services.....	70
4.3 Limitations within the application	71
4.3.1 Dropdown Selection Errors.....	71
4.3.2 Issues with Images and Videos	71
4.3.3 Access and Security Issues	72
4.3.4 Managing Employee Access.....	72
4.3.5 User Adoption Challenges	73
Chapter 5 – Feedback & Review (company)	74
5.1 Feedback from Company	74
5.4 Improvements.....	75
Chapter 6 – Conclusion	79
6.1 Reflection (Individual or Team)	79
6.1.1 Team’s reflection.....	79
6.1.2 Individual reflection.....	79
6.3 Future Prospect.....	84
6.3.1 Safety Checklist.....	85
6.3.2 AI Chatbot (for analytics)	85
6.3.3 Dial Call Function	86
6.3.4 Offline Application.....	87
6.3.5 Language Feature	87
6.3.4 Duplicator Checker.....	87
6.3.5 Enhancement on Filter	88
Appendix/Annexes	89
References (or Bibliography)	96

List of Figures

Figure 1: Approval of Proposal from Company	6
Figure 2: Jotform Software (Formassembly.com,2023)	9
Figure 3: Bubble Software (Bubble,2025)	9
Figure 4: Figma Software (Figma, 2025).....	10
Figure 5: FlutterFlow Software (FlutterFlow.ai,2025)	10
Figure 6: Zapier Software (Zapier.com, 2025).....	11
Figure 7: Appsheet Software (Appsheet,2025)	12
Figure 8: PowerApps Software.....	13
Figure 9: GUI Inspiration.....	17
Figure 10: GUI Designs	18
Figure 11: Reporting Defect Form.....	22
Figure 12: Task View.....	23
Figure 13: Filter Function	25
Figure 14: Filter base on date of creation.....	25
Figure 15: 'Plus' Icon	26
Figure 16: 'Sort' Icon.....	26
Figure 17: 'Reset' Icon.....	26
Figure 18: Task Information.....	28
Figure 19: Task Information.....	29
Figure 20: Dashboard.....	33
Figure 21: Asset & Inventory Main Page.....	34
Figure 22: Asset Management	38
Figure 23: Inventory Management	43
Figure 25: Scheduled Maintenance Calander.....	48
Figure 26: Scheduled Maintenance Form	51
Figure 27: Scheduled Maintenance Details.....	52
Figure 28: Scheduled Maintenance Completion	53
Figure 29: Scheduled Maintenance Reschedule	55
Figure 30: interface panel (Header and Navigation Bar)	57
Figure 31: Push-notification from phone's screen.....	58
Figure 32: Notifications User Experience.....	60
Figure 33: Profile Page.....	62
Figure 34: Help Information.....	65
Figure 35: Home Page.....	66
Figure 36: AI Chatbot	68
Figure 37: Database	69
Figure 38: Task Information (Improved version).....	77
Figure 39: Asset & Inventory (Improved version)	78

List of Tables

Table 1: Stages of Features Development.....4

Table 2: Appsheet vs PowerApps.....14

Table 3 : Defect Reporting Form Questions19

Table 4: Asset Information.....34

Table 5: Inventory Information39

Table 6: Schedule Maintenance Information48

Table 7: Schedule Maintenance Completion Information52

Table 8: Schedule Maintenance Reschedule Information.....54

Chapter 1 – Introduction

1.1 Context of Problem

1.1.1 Initial Problem

Many buildings management team rely on a communal WhatsApp group chat for daily communication and defect reporting. This group chat serves multiple purpose by being the primary platform for cross-departmental communication and all sorts of reporting related to building management. Large number of repetitive defects are being reported through this chat, causing messages to overload which led to the oversight of important messages. Moreover, the management (admins) refer to the chat for defects' information to create work order. Due to the lack of reporting format, admins struggle to process work order whenever there is missing information. This process of reporting and tracking defect has been inefficient for the technicians and management.

1.1.2 Issue Arise by Companies

When proposing the original application to companies, we faced the same question of how this is different from the current software available in the marketplace. This makes us reconsider the objective of the application.

1.2 Objectives of Application

1.2.1 Initial Objectives

After taking in consideration of the problem, we focus this application to streamline defect reporting process with a user-friendly interface. Users like cleaners, technician, and admins can report, resolve, track defects easily. This helps segregates the use of WhatsApp for daily communication and the application for defect reporting. Thus,

reducing the volume of messages sent in the communal WhatsApp group chat, retaining important communication and prevent messages being overlooked.

It consists of defect reporting form, task view and task information. This helps to keep track of all the defects reported and ensure that all the information required are filled.

1.2.2 Final Objectives

After the question imposed by the company, we decided to include other features in the application like schedule maintenance, daily checklist and asset and inventory, making this application focus to streamlining FM operations. This distinguishes our application from those in the market where it focusses on a specific area of FM operations.

1.3 User of the Application

1.3.1 Primary User

The primary target audience of this application are the technicians and cleaners, as they are the frontline in maintaining the buildings. They are expected to report any defects identified and address them promptly. In addition, they are required to update maintenance carried out, routine walkabouts and tracking asset and inventory. This ensures that they are kept accountable of their roles.

1.3.2 Secondary User

The secondary user is the management. They use this application to track technician progress and create work order accordingly. Instead of constantly asking technicians or cleaners of the maintenance works through WhatsApp,

1.4 Proposal for Industrial Linkage

To test that our application would benefit the FM industry, we reached out to our internship companies, Mapletree Investment, Maritime Port Authority and GuocoLand Limited, in hope for their support and the opportunity to collaborate with them. We prepared a proposal to send in an official email to the companies. However, due to company policy and security issues, the partnership fell out. Regardless, this approach allows us to expand our research due to the questions raised by the companies during the initial convincing stage as mentioned in 1.1.2. Following the rejections from our internship companies, we reached out to other possible companies, Property Facility Services Pte Ltd and APM Property Management, with the help from our classmate and facilitator.

1.5 Team Planning

With the project objectives and scope of work identified, we created a schedule that helps us stay on track without feeling overwhelmed by the workload. We decided to create the application in phases. This is to help us cope with the stress managing this project and other modules. We decided that phase 1 would be the following as that is the original objective of our application, streamlining defect reporting system. Once phase 1 has successfully completed, we would start on phase 2. Phase 2 features were created to help improve the efficiency of FM operations and differentiate our application from the current applications available in the market. The fixed features are developed throughout phase 1 and phase 2 as it tends to overlap with the features in both phases.

Table 1: Stages of Features Development

Phases	Phase 1	Phase 2	Phase ½
Features	▪ Defect Reporting Form ▪ Task View ▪ Task Information ▪ Dashboard	▪ Asset & Inventory ▪ Daily Checklist ▪ Schedule Maintenance ▪ AI Chatbot	▪ Notification System ▪ Profile and Setting Page ▪ Help Information ▪ Home Page

Chapter 2 – Research

2.1 22-weeks internship

2.1.1 Origin and Development of Idea

During internship, Li Tong was involved in the defect reporting process where her duty is to report defects and consolidate all reported defects into an Excel spreadsheet for management to track defects more efficiently, which can be tedious and repetitive. This led to the idea of creating an automatic system that updates the database in real time. The process involved various departments such as the cleaners, technicians and admins. Cleaners report defects through WhatsApp chat and provide relevant defect information, while technicians used the details provided to rectify the defects and update the status of work completion. Administrators extract information from the chat to create work orders for defects.

However, the process is unsystematic, as cleaners may provide incorrect or missing information. For instance, forgetting to include the defect location, leading to confusion and inefficiency. Technicians must further enquire the cleaners through the chat, increasing the message volume, while administrators endure the time-consuming task of scrolling through WhatsApp to find the details of reported defects for creating work orders. This leads to frequent oversight of important messages.

2.1.2 Implementation of Solution

To improve the efficiency of this process, Li Tong came up with a solution by creating a Google Form for cleaners to report the defects. The form requires cleaners to provide all necessary information before submission, ensuring comprehensiveness and accuracy with no missing critical information. The submitted data is automatically stored in a

Google Sheet, which displayed all reported defects real-time. Any updates can be instantly viewed by technicians and admins. The defect information is clearly presented, enabling technicians to identify and address defects more efficiently, while admin could create work order more conveniently without any oversight.

The proposed system underwent a one-week trial run after receiving approval from Li Tong's mentor. During the trial, Li Tong monitored the system's performance and gathered feedback from the cleaners and technicians for further improvements. Following the successful trial, management was on board to implement the system until the cleaning contract expired.

4. PowerPoint guide & instructions

- Briefing to use the form (Video to use the form & PowerPoint guidelines)
- Maintenance of Toilet Defect System (PowerPoint to maintain both form & database)

Trial Period: 1/8/2024 to 7/8/2024

During this trial period, the system will be implemented with continuous monitoring to ensure its operation aligns smoothly with the planned objectives. The form and database will be subject to ongoing revisions throughout this timeframe to enhance convenience and functionality.

Actual Implementation: 8/8/2024 onwards, the Toilet Defect Reporting system will be officially implemented, replacing the previous 3-in-1 WhatsApp group chat used for reporting. The system is not to be disclosed publicly, and the prototype cannot be brought back to school. Instead, a video will be provided solely for the purpose of demonstrating how the system works.¹²²

To verify the implementation of the system, kindly request your endorsement on this document. Your signature will serve as confirmation of the system's deployment. Should you need any additional information or have any questions, please do not hesitate to reach out. Your cooperation and assistance in this matter is greatly appreciated.

appreciated. To confirm your acknowledgment and agreement, please provide your name and signature below:

Proposed by: Yip Li Tong (P2225546) Class: DFM/FT/3A/02
Date: 8/8/2024

Name of person in-charge: Senia

Signature & Date: [Signature] 8/8/24

Company Stamp:

Figure 1: Approval of Proposal from Company

2.2 Optimising Solution

As a team, we recognised the potential to further streamline the process by combining the Google Form and Google Sheet into an application. This seamless application would

enhance the defect reporting and tracking process by incorporating all necessary functions, reporting, viewing, updating, and tracking defects within a single organise platform. It streamlines the defect reporting process further by including additional features like filtering and notification. This enhances the efficiency in rectifying defects, increases the overall productivity, and providing greater convenience for management.

Additionally, developing an application enables the team to include other features that can enhance the FM operations. This allows the solution to be a versatile and comprehensive tool with multiple purpose and benefit.

2.3 Considerations for Application Development

2.3.1 Key Considerations in Selecting an App-Building Platform

When evaluating app-building platforms, we considered the following factors:

- a) **Reliability of the Platform** – We evaluate the credibility of the app-building platform and its capability to deliver consistent performance. These platforms must be recognised with proven records of technical and security protocols and stability, such as Google or Microsoft. By using a reliable platform, it reduces risks of technical issues like bugs and errors, preventing disruptions and downtime that could affect the performance of the app. We evaluate this by checking the background of the platform and reviewing through feedback and review from users of the platform.
- b) **Costs and Price Plans** – The costs associated with an app-building platform is critical for budgeting and financial planning. This involves evaluating whether the platform offers free trials, which allow the team to test the features available before committing to a paid plan. Furthermore, we also need to consider the additional costs

for features like publishing the application, additional storage or premium features. Thus, we looked through the plans and pricing offered by the platforms for the best possible option.

- c) **Customisability** – Customisation allows personalisation of application based on the requirements of the company. The overall aesthetic and appearance of the interface is extremely important to ensure the focus is highlighted and visible just by a glance. Therefore, having customisability, provides the flexibility in making necessary adjustments to our liking which helps in harmonising the application to suit the type of theme or aesthetic or even enhance the overall user experience.

Availability of Guides and Tutorials – This refers to the number of guides and learning resources that are available online, which is crucial for us as FM students with no IT background. These resources can be in the form of online videos, articles, and books. It should cover a step-by-step instruction on developing an application, as well as explaining possible reasons and solutions to errors that may arise.

2.3.2 Research on App building platforms

At the start, we explored various building platforms such as Jotform, Figma, Zappier, FlutterFlow, Bubble, Appsheet and PowerApps.

Jotform & Bubble: The elements are pre-made for user to add into application's interface, making it easy to build a simple application. App builders only require dragging the readily available elements into the app's interface, the elements will automatically organise and position themselves to its respective positions. Due to the simplicity of the software, this limits the ability for customisation.

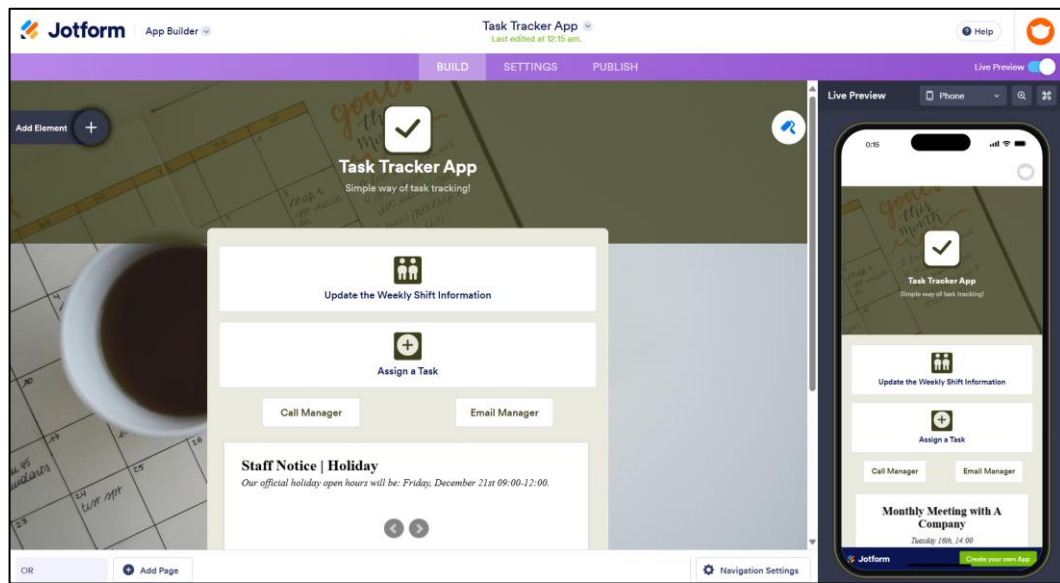


Figure 2: Jotform Software (Formassembly.com,2023)

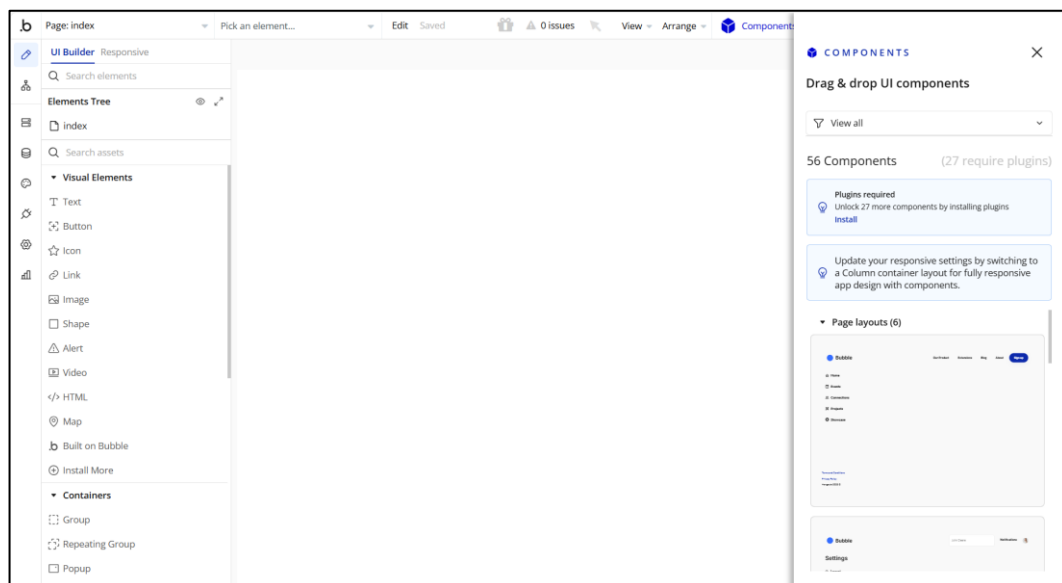


Figure 3: Bubble Software (Bubble,2025)

Figma & FlutterFlow: These complex app building platforms require proficient coding skills, from design layout to the contents and functionality of the application. This offers unlimited possibility to customise the element features to the user's need. However, lacking the coding knowledge, it hinders us from using these platforms.

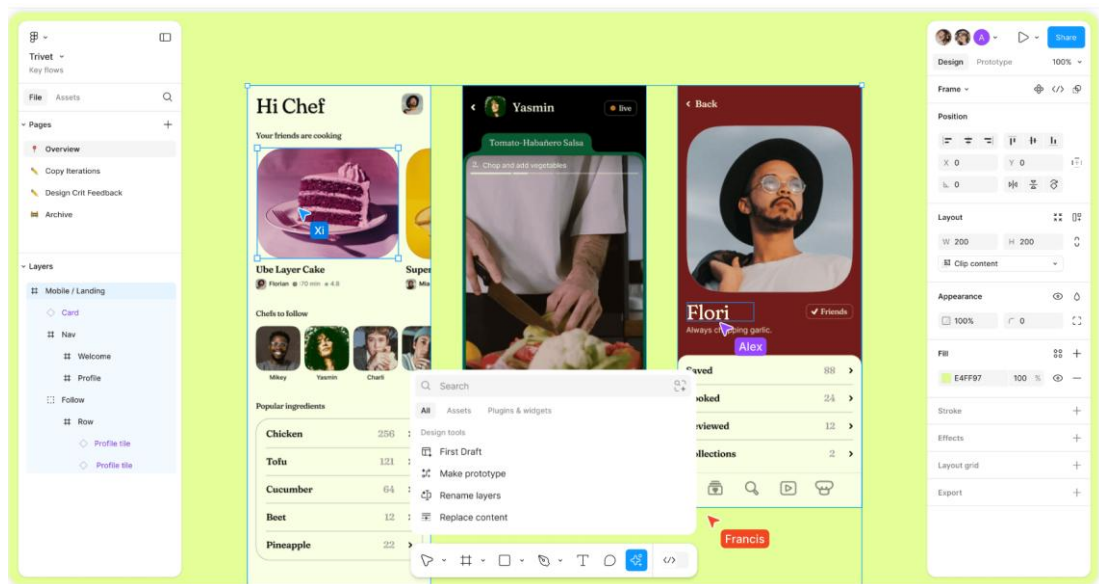


Figure 4: Figma Software (Figma, 2025)

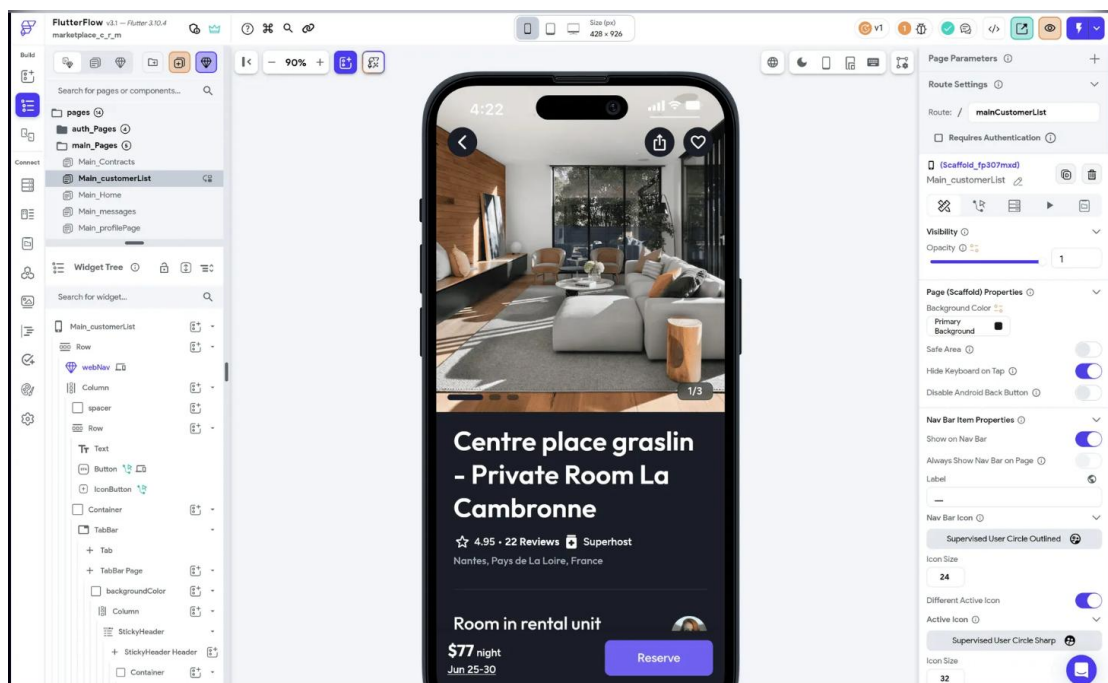


Figure 5: FlutterFlow Software (FlutterFlow.io, 2025)

Zapier: It provides services that allows integration of different platforms into one combined application. Its high cost is resonating with its complexity of integrating

diverse software from different platforms. However, its costly plans and subscription fee has compromised the appeal to us.

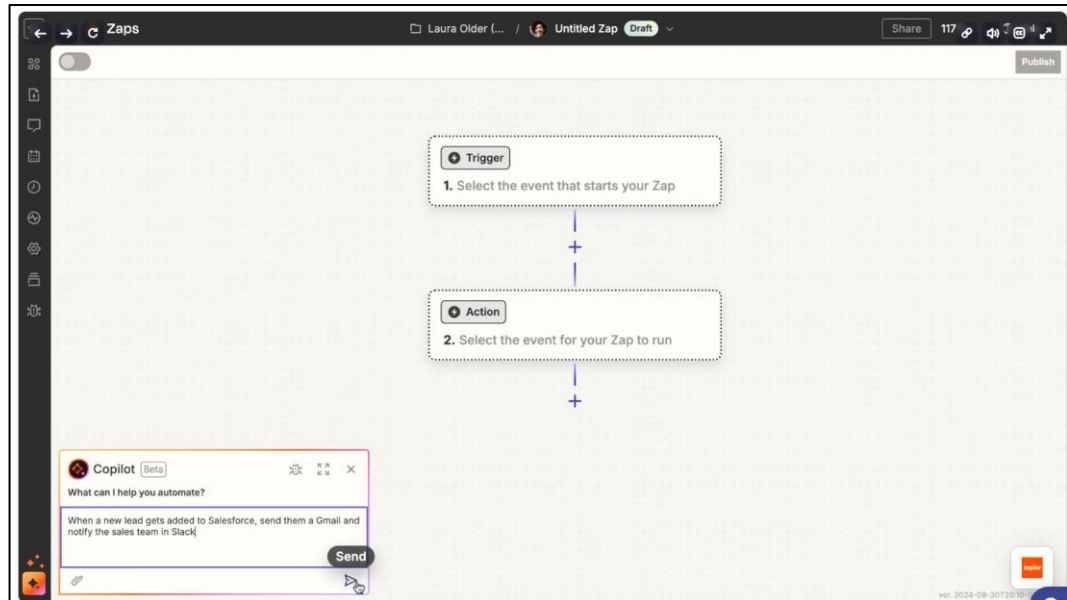


Figure 6: Zapier Software (Zapier.com, 2025)

Appsheet: It is a trusted brand powered by Google, and helpful AI assistance on its right navigation bar to help us with coding. When trying on Appsheet, we realised the layout was disorganised and confusing. Appsheet has a tree view on its left pane, which displays all pages created in the App. User can create, edit, and link data to the app from various format such as Google sheet, .csv, and queries etc.

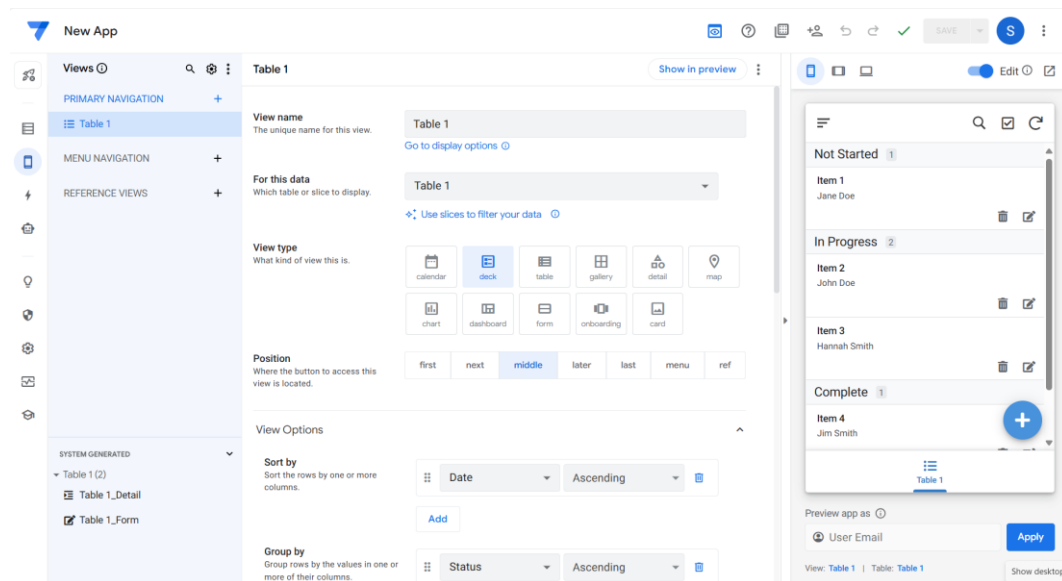


Figure 7: Appsheet Software (Appsheet,2025)

PowerApps: PowerApps functionality is generally quite similar to Appsheet. There is a tree view on that allows user to organise the order of the screen pages with an additional organisation of elements within the screen being categorised. User can also link data from Excel spreadsheet, .csv, and SharePoint etc. However, we discovered that Microsoft had many interlinked software that are more advanced with greater capabilities. Compared to Figma & FlutterFlow, PowerApps is a low-code application building platform, suitable for beginner developers. Like Jotform and Bubble, it has a clear and organised layout, with features pre-made and a developer's interface that can be easily understood.

PowerApps is able to stand out from those two software as it is features' elements are customisable, allowing us to design the application based on our requirements. With Power App being part of the subscription sponsored by SP, we were able to utilise the readily available functions and modern features without any premium purchases. This allows us to explore, develop and program a simple, completed and fully functionable application that helps us achieve our objectivities and scope of work. There are also many resources available online to guide us on the coding required.

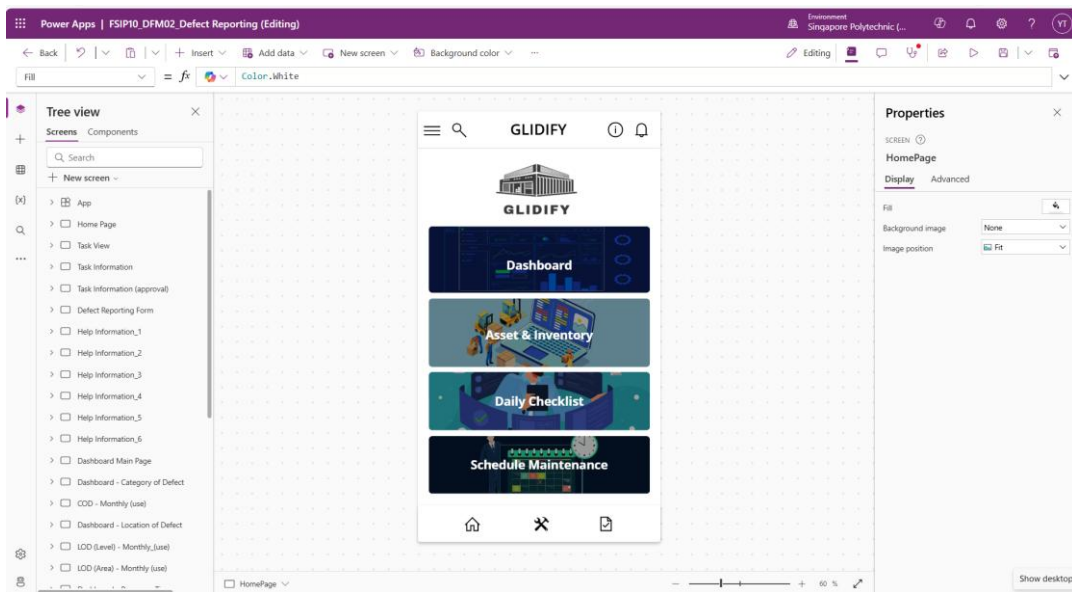


Figure 8: PowerApps Software

2.3.3 Refining Choices

The remaining application building platforms stated in 2.3.3 are excluded from our consideration, as they require subscription for regular use to most of the features. This result in the last two option, Appsheet and PowerApps. Initially, we intended to use AppSheet to build the application. However, as we had no prior experience with app development, we struggled to understand its basic functionalities. While exploring and learning AppSheet, we discovered the PowerApps platform and consider both platforms for review.

2.3.4 Comparison between AppSheet and PowerApps

Both of Appsheet and PowerApps are established brands renowned for their expertise in internet and digital services. However, there are some differences between the two platforms. Each has its own advantages and disadvantages, and their suitability depends on the requirements of the project.

Table 2: Appsheet vs PowerApps

	Advantages	Disadvantages
Appsheet	▪ Powered by Google, making it potentially easier to integrate with original solution, Google Form and Google Sheet ▪ Features clear and aesthetically appealing user interface ▪ Consists of ready-made features for user interface design, requiring only configuration and backend coding or formulas	▪ Limited customisability for instance, the height and position of text boxes are fixed ▪ Little tutorial videos and guides available online ▪ Configurations and formulas are difficult to understand for beginners ▪ Errors in AppSheet cannot be undone or reversed, requiring the creation of a new application from scratch ▪ Does not allow sharing of collaboration links with other users
PowerApps	▪ Offers a wider range of features and elements, each with customisable attributes such as size, position, colour, transparency, font type, and border thickness ▪ Utilises formulas that are easier to understand, with a built-in	▪ Features are not fixed, which may require additional adjustments during development ▪ Being a Microsoft-based software, it is unable to integrate with original solution which is made from Google Forms and Google Sheets

	Copilot feature that explains errors and provides suggestions for improvement ▪ Has an extensive library of online tutorials and guides, teaching users how to create features such as dashboards, forms, and gallery views ▪ Supports collaborative work by allowing users to share links and work together on the application in real time	▪ Certain access requires further subscription which limit certain functionality
--	--	--

Based on the above considerations, PowerApps emerged as the more suitable platform for developing our application due to its customisability, user-friendly features, and extensive support resources. Despite some integration challenges with Google-based solution and minor feature limitation, it provides alternative solutions to meet our requirements. Thus, choosing PowerApps is the best choice for our application as it aligns with our goal to develop an application that streamline FM operations.

2.4 Additional Research

In the early stages of our development, we focused on identifying the features to include in our application. We narrowed down our target group to specify who this application should be for. This way, it helps us identify the crucial features and design that meets their needs in FM Operations.

2.4.1 Research on Application Features

Based on our own experience working in commercial buildings and office towers it sets the foundation for us to begin researching on the arising issues we notice while working onsite. In addition to our understanding of the software available in the market, it set us on the right track as we plan for the development. We considered topics related to how we could create an improvise version of the existing market applications we found. Through our research and experience, we further develop the idea to combine the unique features which resolve the issue raise by companies.

2.4.2 Research on Defect Form Questions

Next, we research on the required information in the defect reporting form. This helps us identify the critical question. We also explore the types of questions suitable as we create the reporting forms.

2.4.3 Research on Coding

Due to the lack of knowledge on computing, we researched numerous Power App programming videos that teaches or guide us on ways to create the codes that we want to input into our feature. Aside from videos, we also looked through electronic books and utilise digital search tools to assist us in coding the application. This is a crucial step that was tedious but extremely beneficial as it has helped us solved countless of coding mistakes and provided us more samples which aid in our development of applications.
(refer to Reference for further information)

2.4.4 Research on Graphical User Interface (GUI)

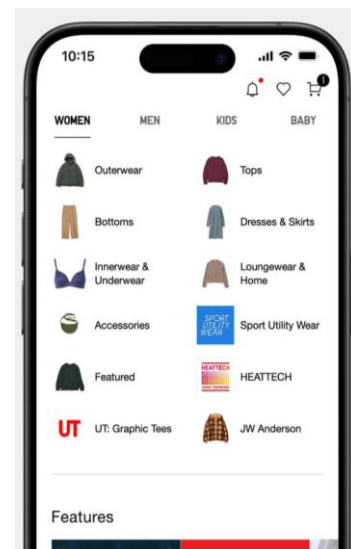
In terms of the GUI, there were several attempts of redesigning and improvement work done throughout each stage to get the version based on our perspective. We mainly take reference and inspiration from our research through apps such as Instagram, Fashion & Clothing online websites and application and a variety of others. We then combined them to re-create into our own version that suits our users. After rounds of researching and re-creating, the team have concluded that simplicity is the best especially with the application catering to a diverse group of ages. With the significance amount of information provided, having vivid colours or icons of large sizes would not be appealing. Thus, we kept it as simple as possible, following the theme of minimalistic and clean aesthetic. This helps us to draw the user's attention to the specific areas we want, avoiding any forms of misleading.



Instagram

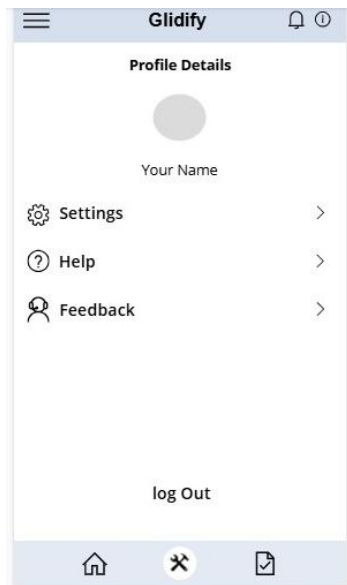


Scaling of the icons

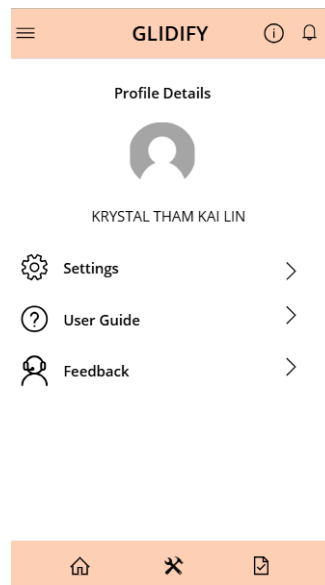


Uniqlo appearance

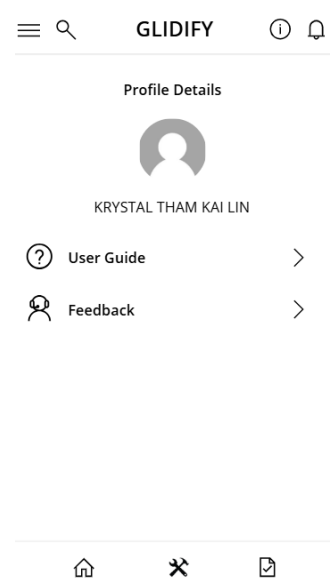
Figure 9: GUI Inspiration



Design Draft 1



Design Draft 2



Final Design

Figure 10: GUI Designs

Chapter 3 – Features

3.1 Phase 1

In this stage, it focuses on the initial objective of the application. It consists of Defect Reporting Form, Task View, Task Information and Dashboard was created.

3.1.1 Defect Reporting Form

The defect reporting form is the first step to create the defect reporting process. User will have to input the defect information manually through the prompted question created by us. The questioned are designed based on the research we have found during the early stage of planning. It contains a series of open and close-ended questions that are related to the defect information, building information such as the specific area and location with photo attachments for visualisation. All questions are required to be filled before submitting to avoid any missing information. This will benefit the technicians when they need to rectify the defects as they have all the required information. All information reported is recorded and stored in a SharePoint database which can be easily retrieve.

Table 3 : Defect Reporting Form Questions

Question	Type of Question	Description
Name	Text	This is to keep track of who is regularly doing their checks for any defects to report. The name is set to automatically appear the name of reporter. This is to prevent them from pretending to be someone else.
Reporting Department	Dropdown	This consists of 4 options, cleaner, technician, security and management.

Building Name	Dropdown	This is prepared if companies have many different buildings or towers. They will be required to pick the building name that the defect is at.
Level	Dropdown	This selection is dropdown to avoid user from entering different types of number to indicate the level like L1 and Level 1. This constant option is crucial when creating a summary report later.
Area	Dropdown	This consists of a list of common areas and the areas segregated within the building. Similar to level, the dropdown option is to avoid having multiple types of writings for the same area. ▪ Toilet (F) – [‘F’ Stands for female] ▪ Toilet (M) – [‘M’ Stands for female] ▪ Lift Lobby ▪ Staircase ▪ Common Corridor ▪ M&E service room ▪ AHU room ▪ Office unit
Note: Building Name, Level and Area is split to 3 different questions in dropdown instead of 1 question answer in a single line text to avoid reporter from forgetting to include a certain information.		
Priority	Dropdown	This focuses on 3 main categories of urgency. Based on the urgency, a set amount of time will be given for technicians to rectify the defect.

		▪ Critical, completed within an hour ▪ Major, completed within a day ▪ Minor, completed within 3 days
Due Date	Time	This can be adjusted by reporter if required otherwise, by default the due date is set based on the priority status with no downtime
Category of Defect	Dropdown	This selection is for user to report based on the category. This assist technician to filter all issues within the same category and rectify at once. In addition, this information is included in the summary report which can help management plan future operations and budget ▪ Cleanliness & Pest Control ▪ Electrical & Lighting ▪ Air Conditioning ▪ Plumbing & Sanitary ▪ Scratches or Damages
Defective Item	Single Line Text	This is simply to state the defective item with no explanation
Defective Description	Multiple Line Text	Reporters are to describe the defective item as detailed as possible for technicians to know exactly the issue to rectify
Photo of Defect	Photo	This is to provide visual aid for technicians to rectify the issue. In addition, this helps to check that the defect is rectify by comparing the before and after photo

The image displays two mobile app screens for the 'GLIDIFY' application, both showing a 'Defect Reporting Form'. The left screen shows the form with the following fields: Name (KRYSTAL THAM KAI LIN), Reporting Department (Which department are you from?), Building Name (Which building?), Level (E.g. for Basement 1, enter '-1'), Area (Which specific area of the level?), Priority (How urgent?), and Due Date (1/14/2025). The right screen shows the form with the following fields: Category of Defect (What type of defect?), Defective Item (Which item is spoiled?), Defect Description (Please describe what is the issue in details), and a Photo of Defect section with a 'Tap or click to add a picture' prompt and a 'Submit' button.

Figure 11: Reporting Defect Form

3.1.2 Task View

The Task View is the main page when user first enter the application. It displays all the defects reported in the form of listing. It would give a brief preview consisting of Priority, Category of Defect, Status and Due Date of the reported defect. The status will be in green if work is completed, and the due date will appear in red if the work has past the due date. This preview provides the main and necessary information for users to understand the basic information about the reported defect. It is kept simple to prevent any confusion and the design clutter-free. For more information on the reported defect, user can click on the defect, and they will be redirected to the Task Information (refer to 3.1.3).

In addition, all reported defects will not be reflected on the user interface forever. We have programmed the preview page to remove the details once it has been 30 days since it is completed to prevent overloading of information in the application. However, all data will still be stored in the Database. Thus, if management requires to review details from the past, they can refer to the Database for all the defects reported with its information. Alternatively, they could the Dashboard Summary to retrieve summarised monthly data. (refer to 3.1.4 Dashboard).

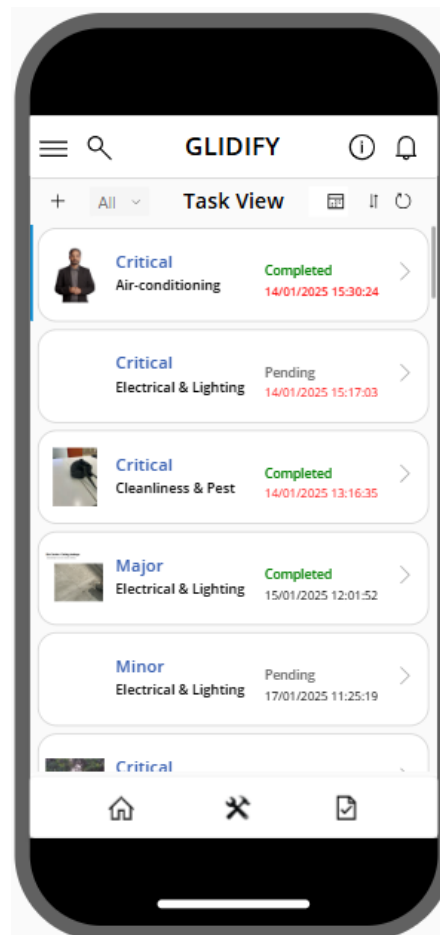


Figure 12: Task View

On top of that, there is different icons aligned with the Task View represents different functions. It assists users to rectify and track defects. It consists of two types of filters, 'Plus' icon to report new defects, 'up and down arrow' to sort and a 'refresh' icon to reset the task view.

Filters:

The filters can help users can choose to reduce the time spend on searching for the reported defect details. The Main filter function is through the dropdown list. It is programmed to filter the data base on the 'Priority' and 'Status of work' as shown in Figure 13. By selecting any of the item listed in the dropdown, the user interface updates automatically accordingly. For example, by ticking 'Pending', it would reflect all the defects with the status, pending. One unique factor is that the filter function is a multi-checkbox feature, where user can select multiple items according to their preference or needs.

The calendar filter function is programmed to filter according to the created date of the reported defect. For instance, if the defect was reported on 23rd January 2025. User can filter the dates by tapping on 23rd January 2025 and it will update the data to show all the defects reported up to 23rd January 2025.

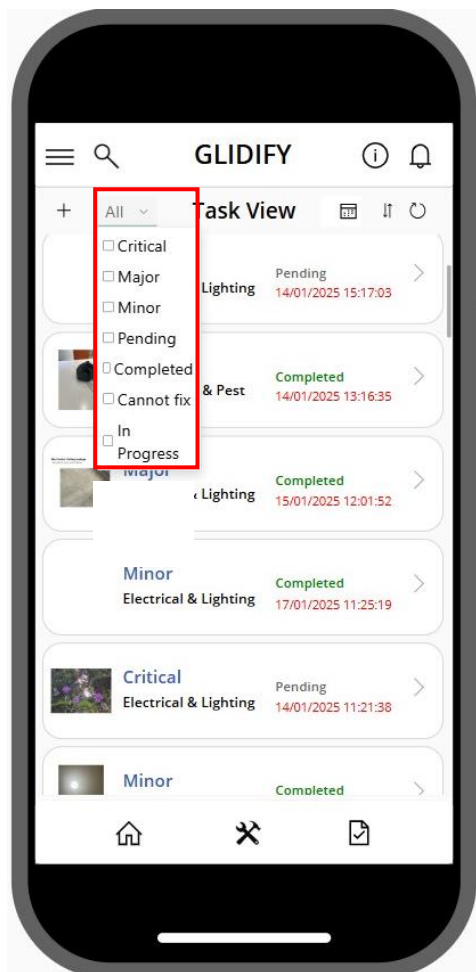


Figure 13: Filter Function

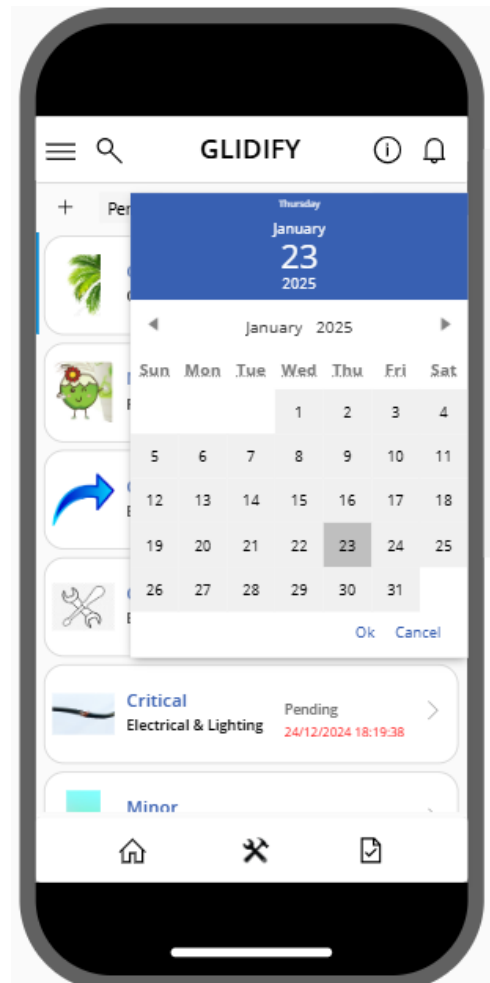


Figure 14: Filter base on date of creation

‘Plus’ Icon Feature:

The ‘Plus’ icon next to the Main filter feature is programmed to direct users to the Defect Reporting Form. Having a variety of methods to retrieve the same information, provides versatility and makes the app more user friendly.

Sort Feature:

The Task View can also be sorted using the ‘up and down arrow’ next to the filter calendar which is programmed to rearrange the task view in either last created or first created .

Sort Feature:

The Task View can also be sorted using the ‘up and down arrow’ next to the filter calendar which is programmed to rearrange the task view in either last created or first created .

Reset Feature:

The refresh icon is programmed to reset the task view back to its original state. This is extremely useful when users want to reset their task view without having to remove all the filters one by one.

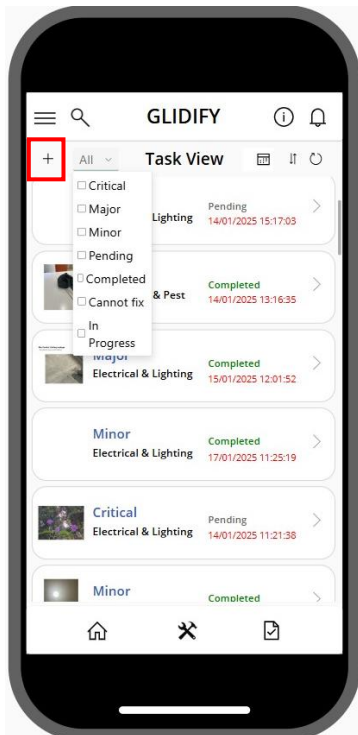


Figure 15: ‘Plus’ Icon

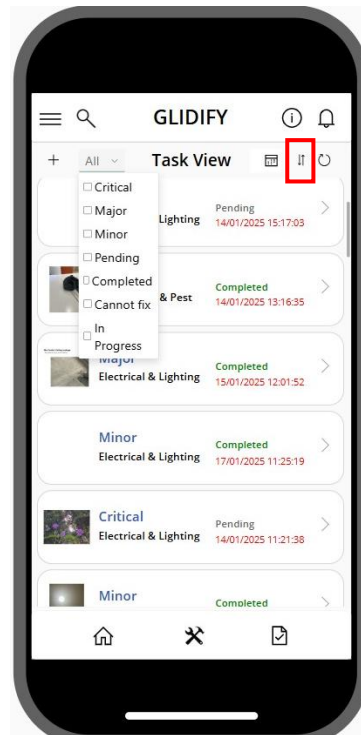


Figure 16: ‘Sort’ Icon

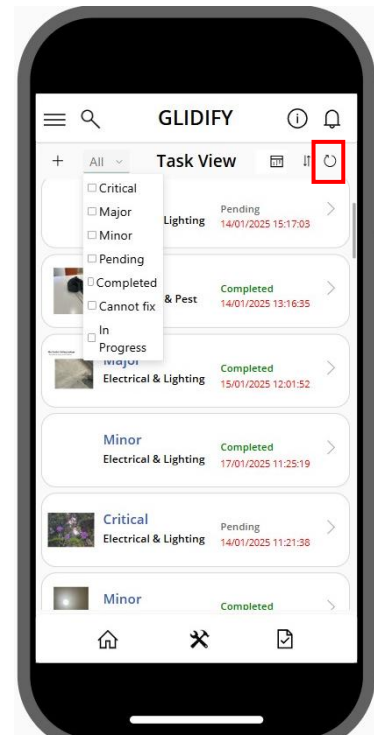


Figure 17: ‘Reset’ Icon

3.1.3 Task Information

The task information is used for both reviewing the defect details as well as updating the work status. To reduce the need to tap multiple times to be redirected, we included both sections into one full experience. On top of that, all the updated data are stores in the same database SharePoint for easy referencing.

Fix Details

Details that were filled up previously when reporting the defective item, are known as Fix Details as they are unalterable. This prevents technician from altering the information. It serves as an informative section for technician to refer to before or when they are fixing the issues reported through the application. Once the work is completed, they are required to scroll down to fill in the updated details base on the questions provided.

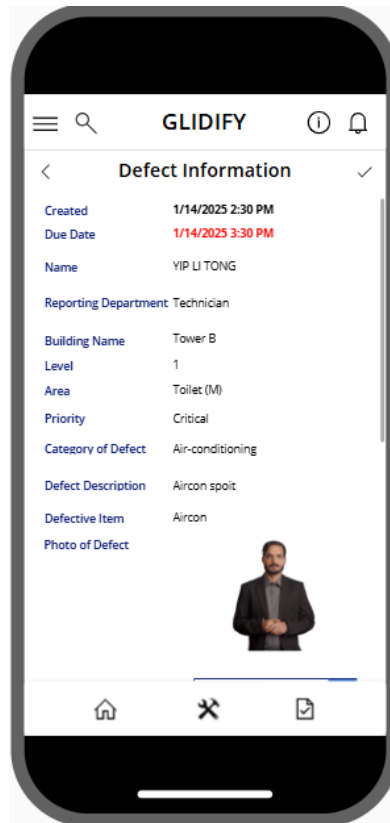


Figure 18: Task Information

Defect Update Details

There are three questions to close this defect report case which we believe is essential to ensure that users performing the task completes it with quality and takes necessary accountability of their actions. This helps the management to check if the defects have been rectified by comparing the detail when first report and updated details. Thus, the management would not require to physically check all the defects which can be time consuming.

1. Working Progress. Once the work is completed, user is to select the 'Complete' choice from the drop-down menu. In any case if the work is unable to be completed due to certain reasons, they are to select the 'Cannot Fix' choice which will reduce confusion if the work is pending for a prolonged amount of time. This helps to keep track of the defect status.

2. Photo of Completion: This is to showcase the after image of the defect as evidence that the work is completed properly.
3. Remarks: This is not a requirement, but if user were to have any remarks to comment on the defect, they may do so to fill it up. However, in any case if the work is marked as 'Cannot fix' it is advisable to include remarks to provide proper reasoning to prevent confusion.

However, after reviewing the application with our industrial linkage. We have readjusted the questions in this section to further enhance and ensure proper standard of rectification is met (**refer to Feedback & Improvements**).

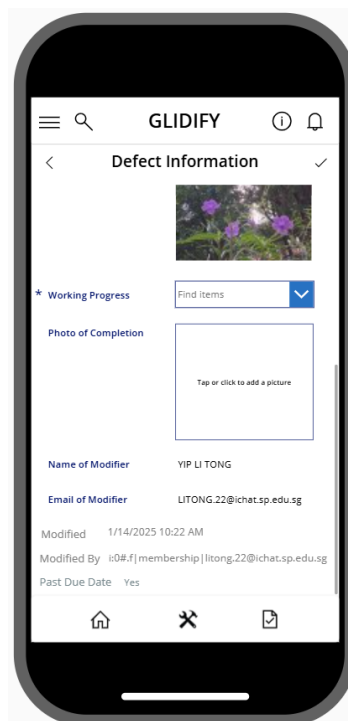


Figure 19: Task Information

3.1.4 Dashboard

Apart from the defect reporting form, the dashboard is what rounds up the whole cycle of reporting. In FM industry, it is utterly important to provide detailed breakdown of monthly, quarterly or yearly work to make data analysis easier and efficient for meetings.

Taking this into consideration, our team have created a summary dashboard which compiles monthly data into graphs or charts to highlight the areas that are important to the management. Based on our understanding and experience in internship, we have come up with three major categories which consolidate data from the Defect Reporting Database monthly.

- 1) **Categories of Defect:** The categories of defects measure the occurrence of the type of defect that is reported continuously throughout the scale of four weeks. For this category, we have chosen Bar graph as our selected experience as it provides a simple interface for user to read at one glance and compare the number of defects based on the category. This helps management to plan the budget required for each category based on the frequency of occurrence of the defect category.
- 2) **Location of Defect:** The Location of Defect comprises of data collected based on the most reported location from the Defect Reporting database. The intended sub-categories our team wanted to provide was categorised based on Building Level and Area. However, due to time constraints, we only managed to complete the Building Level. Nevertheless, the data is presented in pie chart as it is according to the different levels in the building tower. This help management to monitor the levels with arising issues especially if they have tenants in the building.

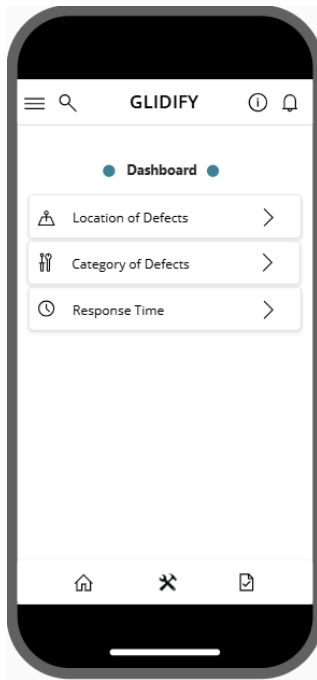
3) **Response Time:** This category focuses on the speed of completion of the reported defects which are all presented in pie charts. Categorized into three sub-portions, Priority Breakdown, Unsolved Cases and Surpass Duration.

Priority Breakdown focuses on the number of priority cases based on the selection among Critical, Major or Minor. This would be vital when management to plan mitigation measures and allocation of resources.

Unsolved cases present the total number of unrectified cases for the month. Based on the summary report, it will show whether the technicians are able to complete the task and if there are any oversight of tasks that were not completed.

Surpass Duration involved the total number of cases that were completed late for that entire month. This data can be used as evidence for their Key Performance Index or even the overall teamwork and communication of the management.

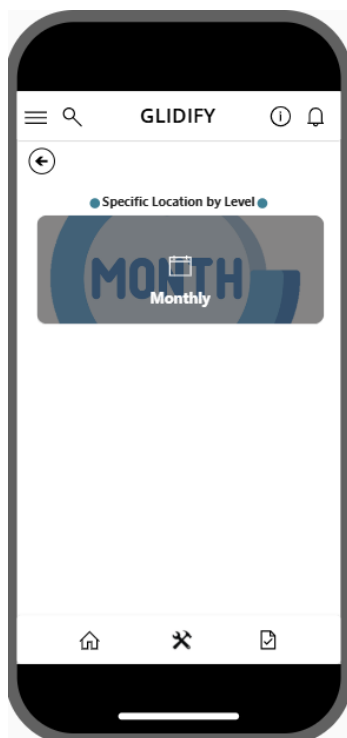
Overall, the categories can be customised to fit the type of building or request the management want from us. By focusing on these three factors, we believe that it would provide general if not in-depth summary of all their work done during each month. In addition, it also saves trouble for the management to manual collect all the data and create excel sheets which is not as feasible for presenting purposes.



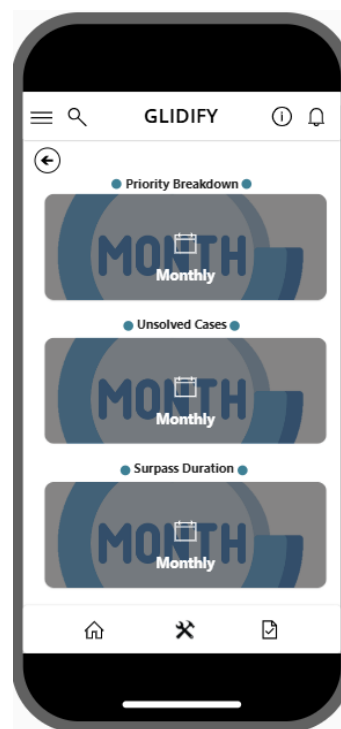
Dashboard Main Page



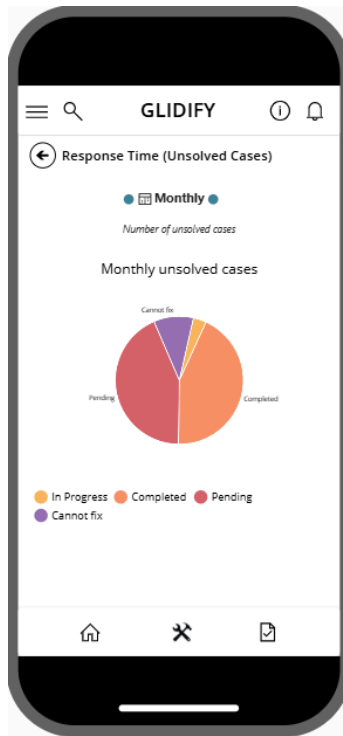
Category of Defect



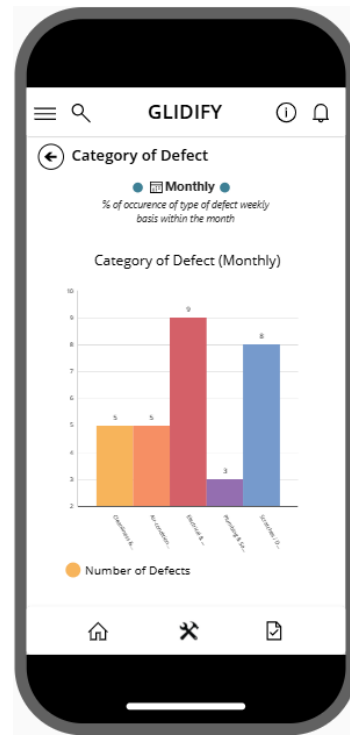
Location of Defect



Response Time



Pie Chart



Bar Graph

Figure 20: Dashboard

3.2 Phase 2

Phase 2 is carried out after the completion of Phase 1. Phase 2 features the upgrading of the application by including functions that could streamline the FM operations, our final objective.

3.2.1 Asset & Inventory

Taking inspiration from our internship experience, we recognised the importance of consistently tracking Assets & Inventory. This is to keep the operations team accountable on the usage especially when updating this information to the upper management. Having a feature to help the management to keep track of their Assets & Inventory would reduce time spend to consolidate every furniture or light bulbs that are maintained by them. This would be vital especially during annual spring cleaning or when auditing.

Weighing the types of categories we believe is important for the management, we have come up with several sub-elements that fulfil the overall idea of our Asset & Inventory.

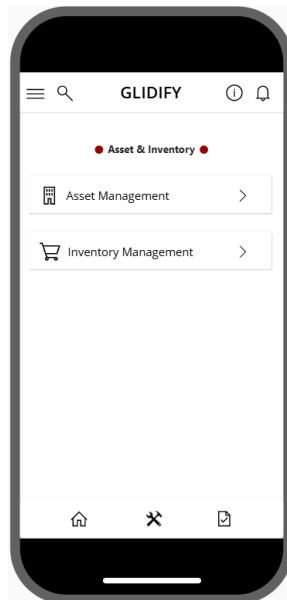


Figure 21: Asset & Inventory Main Page

Assets:

Asset Management, in Business terms refers to the process of managing asset to increase the value of the company, reduce overall cost and risk which involves both tangible and intangible assets. However, in terms of Facilities Management Operations, we focus on only tangible assets that are valuable to the company for instance Heat, Ventilation and Air-Conditioning (HVAC), Office Furniture, Machinery and others. The information listed in Assets are the following

Table 4: Asset Information

Type of Information	Explanation
Title	This is the name or label given to identify the specific asset.

Model	This is the specification of the asset. This provides convenience when purchasing spare parts or replacing the asset. This ensure the purchase of spare parts is compatible for the asset.
Warranty	This shows how many years left that the company could expect maintenance service and replacement service if there any damage. The management will be able to plan the budget to include additional maintenance work or replacement of asset with this information.
Location	The information is crucial to prevent missing or mismanagement of asset, especially for easily moveable assets like projector or TV.
Photo	Photo of assets helps provide visual aid to illustrate the appearance of asset.

For our project, we have come up with four different types of assets that we feel are important and valuable to the company for recording purposes.

1. **HVAC:** We have chosen HVAC as it is one of the most important assets in a building. It provides air ventilation and comfort to occupants. There could be numerous units and types located on each level of a building. Thus, it is important for us to ensure there is a record of the number of units and other information for ease to maintain them. At the same time, this also benefits the contractors who are conducting monthly or quarterly maintenance for the units in the office. There would be no downtime if the contractors require the

asset list to identify the specifications for all the required filters, pipe change etc.

This includes all the Fan Coil Unit (FCU), Water Coil Unit (WCU), Air Handling Unit (AHU) in the office towers that are handled by the management. On top of that, it also includes the Cooling Tower, compressor that are located on the rooftop of the building tower.

2. **Office Furniture:** This asset list is crucial as it includes the whole list of furniture and fittings under the care of the management. By having a consolidated list before hand, it eases the job of the team handling the leasing as they do not need to spend too much time checking if the furniture is missing or being moved. Also, If the management requires to transfer the furniture to a different level or building, they would be able to easily retrieve the total number of furniture and its location. This helps to facilitate the process. Apart from that, it also gives management a sense if there is a need to purchase more or replace office furniture.

For instance, there could be lounge tables and chairs, sofa sets with coffee tables, shelves of all sizes, cushion chairs etc which can be added into the list.

3. **Projector/TV:** This is important because some projectors and television are movable. There are cases where irresponsible users will misplace them after moving the device around the office. Thus, by keeping track of the number of devices there are and having the last modified personnel to fill in the

updated copy, it not only keeps tracks of the total asset but also ensures users take accountability and return the devices to its original place after using.

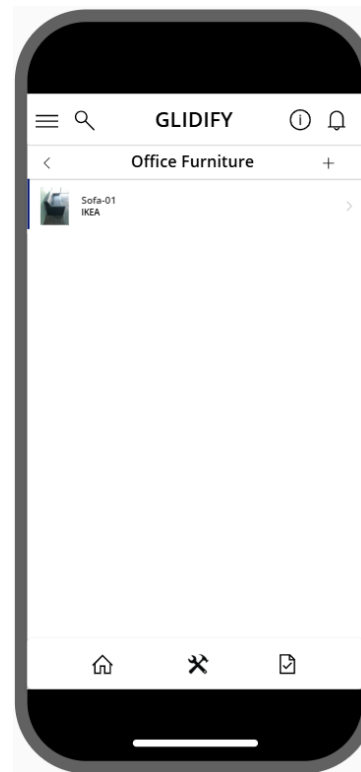
Furthermore, it also provides clarity for the maintenance team. By having the total numbers prepared beforehand, it reduces the need for them to keep confirming on the number of devices they require to provide maintenance for.

4. Others. For instance, lift systems. Although, the lift is stationary, it is also important to keep track of the information to ensure that lift is always working at optimal functionality.

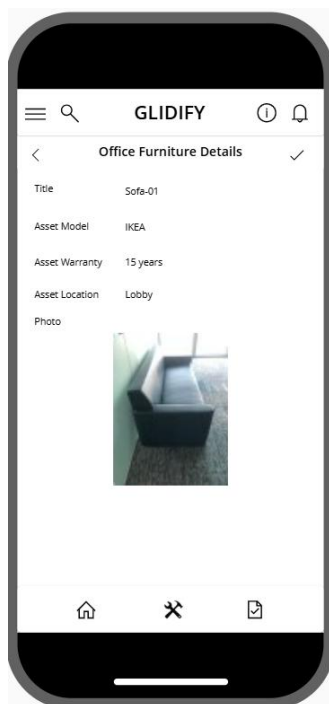
In terms of designing, we decided to keep the user experience simple to emphasise the key assets. To create the Assets listed, user will have to manually input some information based on the prompted questions we have provided. This includes the name and model of the asset, warranty date and its asset location with photo attachment. By tapping on the tick on the top right hand of the experience, the data collected will be stored in our database and reflected on the Asset Listing page for example 'Office Furniture'. Alternatively, they could enter all the information directly in the database and use the form for newly added assets. User will be able to review the details by tapping on list of furniture of their choice and they will be re-directed to the Detail page that reflects the recorded information.



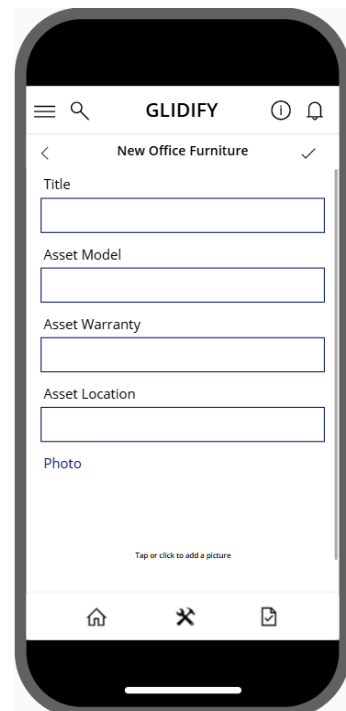
Asset Management Main Page



Office Furniture List



Office Furniture Details



Office Furniture Form

Figure 22: Asset Management

Inventory:

Inventory Management is the process of managing the company inventory to keep track of the stock listing and ordering list. Inventory in Facilities management, involves the items that requires consistent restocking and replacement work. For the list of inventories, we have come up with, we have chosen common items that are frequently used in office towers.

Table 5: Inventory Information

Type of Information	Explanation
Title	This is the name or label given to identify the specific inventory.
Model	This is the specification of the inventory. This provides convenience when purchasing spare parts or replacing the inventory. This ensure the purchase of spare parts is compatible for the inventory.
Warranty	This shows how many years left that the company could expect maintenance service and replacement service if there any damage. The management will be able to plan the budget to include additional maintenance work or replacement of inventory with this information.
Quantity	This represents the total number of inventories there are as of current. The management will be able to use this to determine if new purchases need to be made.

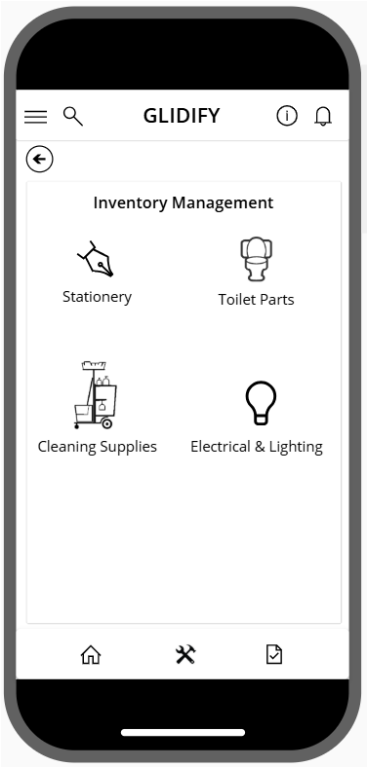
Location	The information is crucial to prevent missing or mismanagement of inventory, especially for small items such as stationaries and individual toilet parts.
Photo	Photo of inventory helps provide visual aid to illustrate the appearance of inventory.

1. **Stationery:** This can be used for the management's office to keep track on the stationery supplies. Base our team's internship experiences, we have realised that stocklist of stationery supplies is usually done by the administration team. To lessen their workload, we created a stationery inventory list to ensure that all employees play their part by updating whenever they take a pen for instance from the stationery shelf.
2. **Cleaning Supplies:** In some cases, the cleaning company is an outsourced service provider. Which means their cleaners are outsourced, but the cleaning supplies are paid by the management the cleaning company works with. This means that it is important for the management to keep track of the cleaning supplies usage to ensure that there is no overspending of expenses. This also help ensure the cleaners are kept accountable of the usage of cleaning supplies.
3. **Toilet Parts:** In an operating building, the cleanliness and appearance of the washroom is extremely important. Therefore, it is recommended to keep proper track of the individual toilet parts such as bidet heads, tap filters, soap dispenser's containers, toilet rolls to prevent unnecessary downtime to wait for the arrival of new parts. Having it tracked and monitored in our application, would allow management

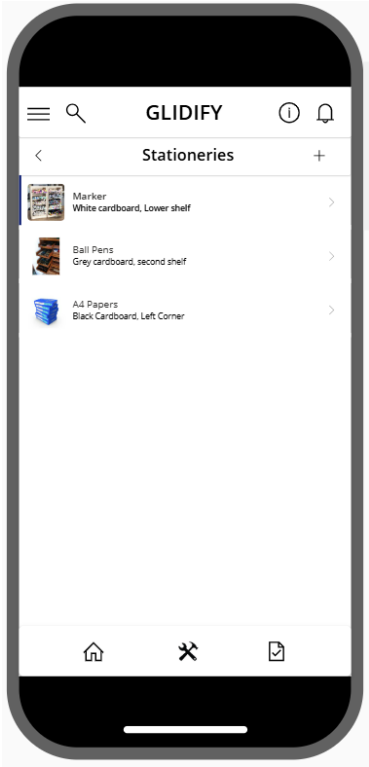
or even cleaners to make quick reference to the amount of stock left and request for new orders when low in stock.

4. **Electrical & Lighting:** To keep an operating building lighted, electricity and light bulbs are necessary. Therefore, it is important to ensure there are enough or extra replacements for bulbs that fussed off due to overheating or wear and tear. Keeping track in our application, gives the management an idea as to whether new stocks need to be ordered. It is also useful whenever the user needs to find a particular model hurriedly and do not have trouble searching as our app presenting the location and number in stock.

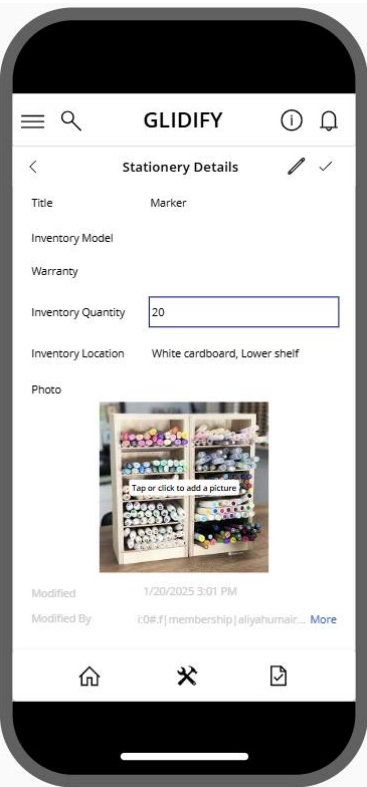
To create the inventory list, user would input similar details as Asset. They will be required to fill in blanks related to the inventory model name, location of the inventory, total current quantity of the inventory and a photo attachment for visualisation purposes. User will be able to review the details through the individual inventory page just by a tap on the screen. For instance, when there is replenishment of stock, user can easily hit the 'pen' icon in the Details page to make amendments to the information. The last modified personnel and last date of modification which will allow management to track the last person who used the inventory.



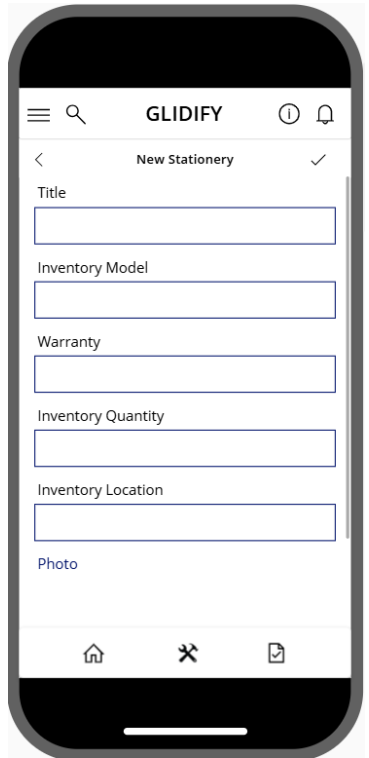
Inventory Management Main Page



Stationeries List



Stationeries Details



Stationeries Form

Figure 23: Inventory Management

3.2.2 Daily checklist

One of the common task technician and cleaners do daily is to conduct walkabout checks all around the premises under their care. This is extremely important as it concerns the cleanliness, appearance, aesthetic and functionality of the building image. Therefore, by having a Daily Checklist it serves as a guideline for users to follow and tick off as they go about conducting their daily checks. All the items are customisable to accommodate to the company's information and facilities they handle. Most companies would have a daily checklist, but it is in the form of hardcopy. This method lack of evidence that the cleaners and technician did conduct the daily walkabout. Thus, they would need to provide a photo as evident of work completed. User will submit the completed checklist for the day which is stored in individual SharePoint database with time it was created for the management to review. The purpose of the evidence system is also to ensure user is responsible for the work done and prevent any unwanted miscommunication if the work is not completed up to standards.

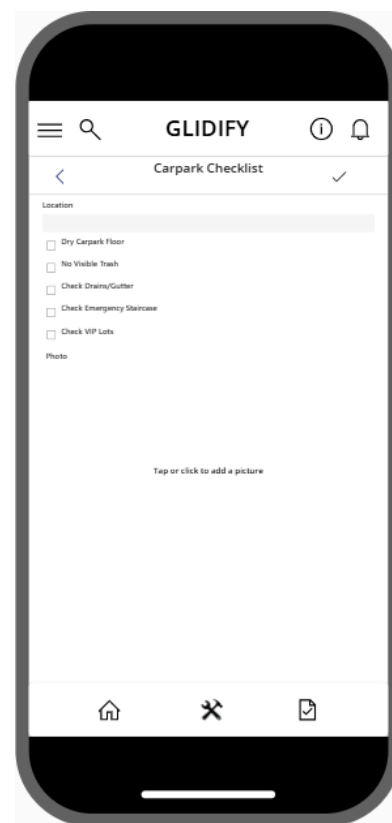
In terms of the functionality of the application, there will be a few categories to fill up which are listed below:

1. **Carpark Checklist:** This checklist comes into play when user conduct cleanliness checks around the carpark premises. This includes some key items such as ensuring that there are no litters around, no wet or car marks on the floor and unauthorised vehicles should not be parked at VIP area.

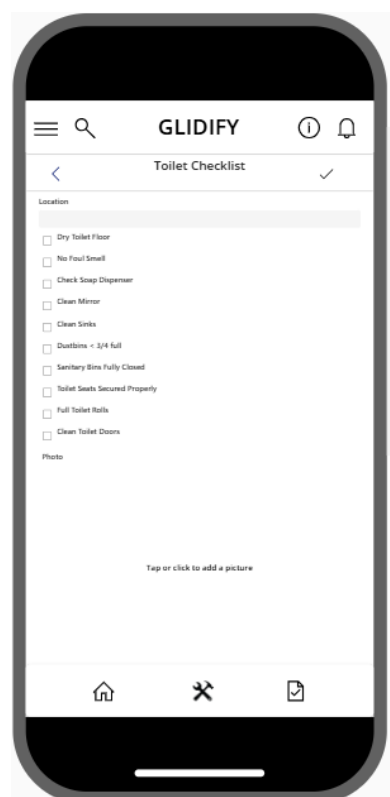
2. **Toilet Checklist:** This includes a variety of items to tick off for instance ensure the soap dispenser is at least three quarters filled, the sinks are clean, no foul smell from floor traps and the toilet rolls in the cubicles should always be refilled.
3. **Lobby/Office Checklist:** This focuses mainly on the appearance and cleanliness. It is more so focus to ensure it maintain the image the office upholds. It is therefore important to ensure that all dustbins are either empty or half-filled to accommodate to the crowd that comes and goes. The frequency of the lift should also be properly measured to ensure that there is no delay in speed especially during peak hours.
4. **Walkabout checklist:** This checklist mainly focuses on the general areas such as the common pathways towards the office towers or shopping mall premises that are handled by the management. It includes general items such as, no stacked rubbish near emergency exits or entrance exits, atrium ceiling lights should not be turned on before a certain timing and no coffee stains should be seen on the floor.



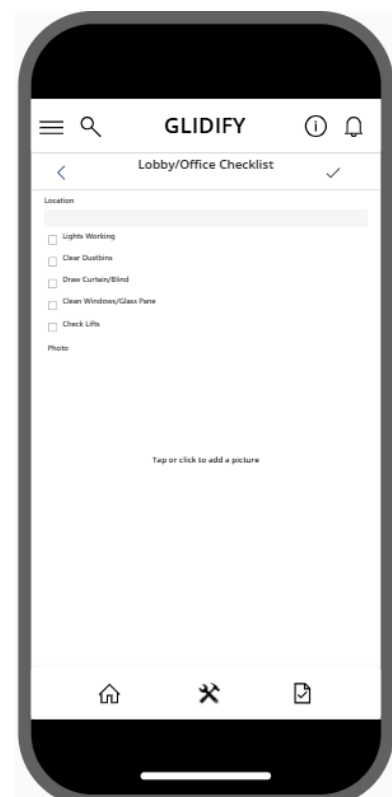
Daily Checklist Main Page



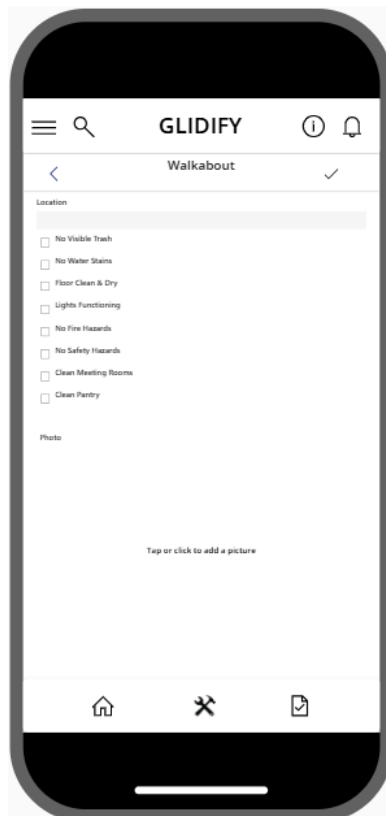
Carpark Checklist



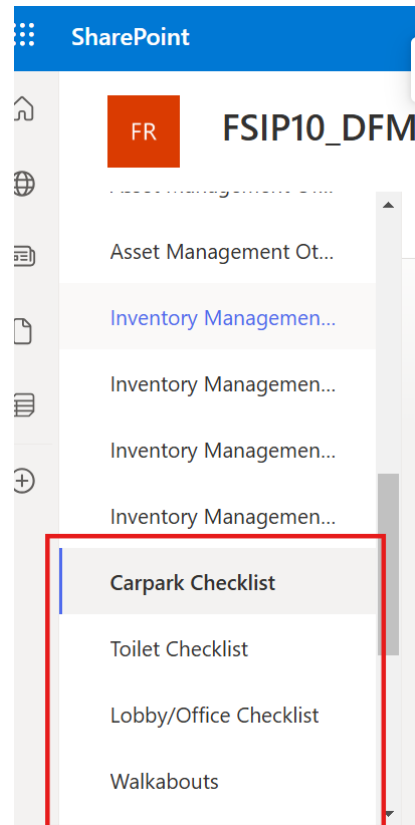
Toilet Checklist



Lobby/Office Checklist



Walkabout Checklist



Individual SharePoint database folder

 A screenshot of a SharePoint list titled 'FSIP10_DFM02_Defect Reporting_From_M365'. The list is named 'Carpark Checklist'. It contains a table with columns: Location, Check Dr..., Check Em..., Check VIP..., Photo, Created, and Created By. The table has five rows of data, including entries for 'Basement 1' and various dates in January.

Location	Check Dr...	Check Em...	Check VIP...	Photo	Created	Created By
Basement 1	✓	✓	✓		January 13	NUR ALIYAH UMAIR
	✓		✓		January 14	KRYSTAL THAM KAI
	✓				January 14	KRYSTAL THAM KAI
	✓	✓			January 14	TAN WEI SHAN
	✓	✓	✓		Yesterday at 10:28 ...	YIP LI TONG

Example of data base

Figure 24: Daily Checklist and forms

3.2.3 Scheduled Maintenance

The scheduled maintenance is a feature used to focus on the different upcoming preventive and corrective maintenance. It is important to ensure that all the required maintenance work are done consistently and with standards. Thus, by having this feature, it keeps tracks of all the necessary work scheduled throughout the year. Ensuing that users will not miss out any important repair or replacement work that could result in serious downtime.

The calendar is designed to make it user friendly, ensures clarity and ease of retrieving details just by a click of the calendar. On top of that, it also includes features for users to ‘delete event’ or ‘reschedule’ accordingly which promotes flexibility and advance functionality. All icons used are interconnected to re-direct users to different pages.

It contains five different screens:

1) Schedule Maintenance Calendar:

In this screen, it includes a calendar interface that allows user to click on specific dates to view basic information such as, Types of Maintenance, Building Location, Start and End time set for the scheduled work. furthermore, by selecting a particular date, if there is scheduled work to be completed on that date, there will be the preview of the event below the calendar for user to tap on to view the event details.

On top of that, on days with scheduled work, there will be a small blue circled programmed to appear below the actual date. This serves as a reminder to the users that they have work to be done.

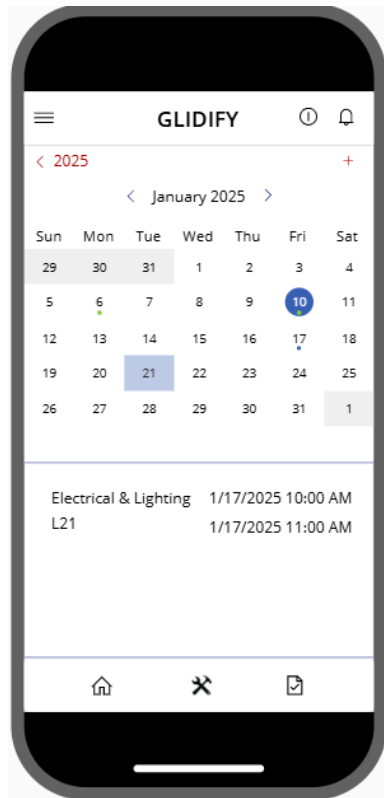


Figure 25: Scheduled Maintenance Calander

2) Creation of Form:

This screen contains a Schedule Maintenance Form that requires users to input their upcoming work according to the requirements. There are eight required items to be filled out before submitting which will be listed below.

Table 6: Schedule Maintenance Information

Question	Question Type	Description
Types of Maintenance	Dropdown	This consists of a list of types of maintenance that requires scheduling of work to be completed. ▪ Electrical & Lighting

		▪ Plumbing & Sanitary ▪ Landscape ▪ HVAC
Building Location	Dropdown	This selection is dropdown to avoid user from entering different types of number to indicate the level like L1 and Level 1. This constant option is crucial when creating a summary report later.
Building Area	Dropdown	This consists of a list of common areas and the areas segregated within the building. Similar to level, the dropdown option is to avoid having multiple types of writings for the same area. ▪ Female Toilet ▪ Pantry ▪ Lobby
Task Description	Text	This section allows users to input an explanation of the schedule work to be completed.
Start date & time	Time	The start date and time refers to the day of the actual work and the timing the work should be carried out. It is important to follow strictly to avoid unnecessary downtime.

		Refer to Table 8 for reschedule if unable to carry out within the stated time.
End date & time	Time	The end date and time refers to the day of the actual work and the timing the work should be carried out. It is important to follow strictly to avoid unnecessary delay for other work on the same day. Refer to Table 8 for reschedule if unable to compete within the stated time.
Before Photos	Picture	This section requires a photo attachment of the activity to provide visualisation for aphantasia users.
Remarks	Text	This section is not required for fill out. However, if user have a comment related that could affect the procedure of the work, they could fill in their remarks under this section.

The image displays two mobile app screens for the 'GLIDIFY' application, both showing the 'New' maintenance form. The top navigation bar includes a menu icon, a search icon, the app name 'GLIDIFY', and an information icon. The left screen shows the form with the following fields: 'Types of Maintenance' (a dropdown menu with 'Find items' selected), 'Building Location' (a text input field), 'Building Area' (a text input field), 'Task Description' (a text input field), 'Starts' (a date and time picker set to 12/31/2001 00:00), and 'Ends' (a date and time picker set to 12/31/2001 00:00). The right screen shows the form with the following fields: 'Starts' (a date and time picker set to 12/31/2001 00:00), 'Ends' (a date and time picker set to 12/31/2001 00:00), 'Before Photo' (a large image placeholder with the text 'Tap or click to add a picture'), and 'Remarks' (a text input field). Both screens have a bottom navigation bar with icons for home, a crossed-out icon, and a document icon.

Figure 26: Scheduled Maintenance Form

3) Event Details

This screen contains a comprehensive detail of the scheduled work. It allows user to view schedule work, area, any remarks to be taken note of without altering any information submitted step 2. User could also tap on 'Delete' to remove the scheduled work. Once the work is completed, user can select to fill in the Completion form by clicking onto the 'Completion icon' on the top right corner and reschedule the maintenance work by clicking onto the 'Pen icon' next to the 'Complete icon'.

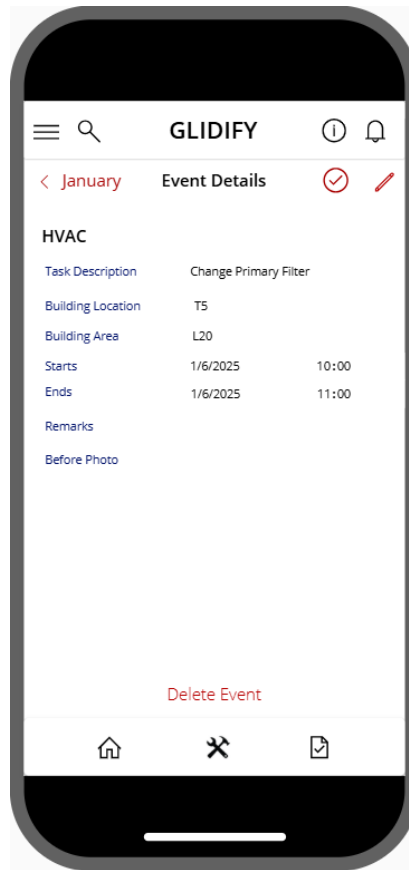


Figure 27: Scheduled Maintenance Details

4) Completion of Form:

Once work is completed, user is to fill this form to confirm the work completion. It includes three new items to be inputted:

Table 7: Schedule Maintenance Completion Information

Question	Question Type	Explanation
Before Photos	Picture	This section is used to show case the condition of the maintained item before maintenance works is carried out
After Photos	Picture	This section is used to show case the maintenance work carried out after the completion of the scheduled maintenance.

Completed By	Text	User completing this form should fill in their name as they are the person who have completed the scheduled work.
Completion Remarks	Text	This section is not required for fill out. However, if user have a comment that is important to note down, it is advised to fill in.

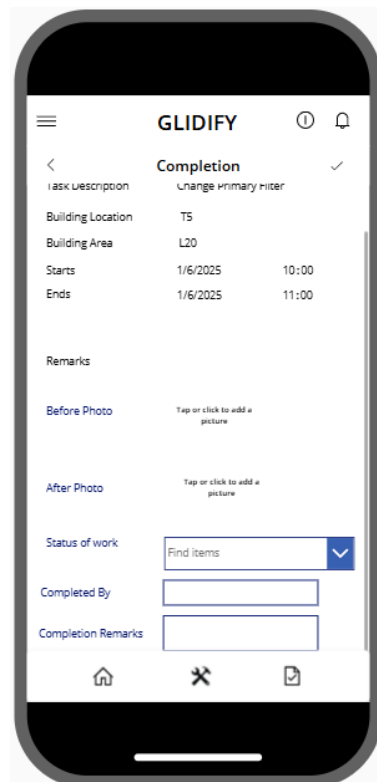


Figure 28: Scheduled Maintenance Completion

5) Rescheduling of Form:

This screen is created to allow flexibility for user to replan their scheduled work in case if they were unable to complete on the actual day. It includes three items to complete the rescheduling procedure:

Table 8: Schedule Maintenance Reschedule Information

Questions	Question Type	Explanation
Reschedule Start date & time	Time	This section is for user to input the new Start time and date to implement the postpone of the work.
Reschedule End date & time	Time	This section is for user to input the new End time and date to implement the postpone of the work.
Reschedule Remarks	Remarks	This section is required to be filled as user must state the reason to why they are unable to carry out the work base on the stated time and date.

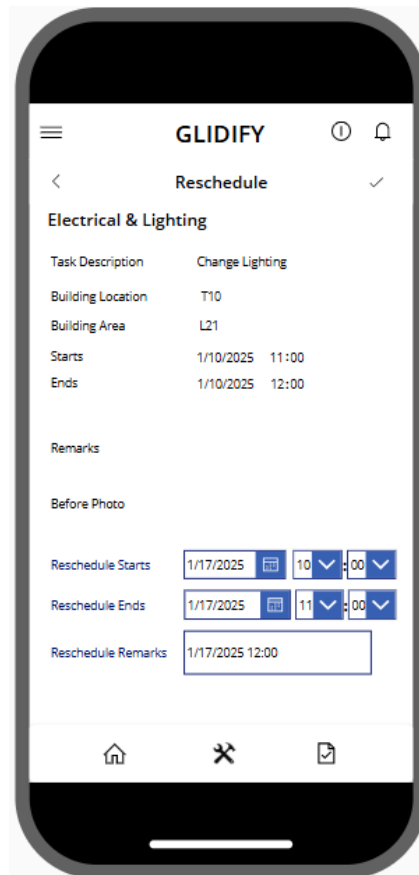


Figure 29: Scheduled Maintenance Reschedule

3.3 Fixed Features

The term Fixed Features represents the items that are always present throughout the entire application. With the thought of ensuring that certain icons such as profile page, notification bells are always reachable with a tap on the screen. We have decided to fix them across the entire application, no matter what screen user is at, they will be able to tap and be re-directed to the page of their choice. It is easy and friendly to users who we believe who prefers to have a simple user experience.

3.3.1 Interface Panel

The interface panel is one of the fixed features presents throughout the entire application. It includes two panels: Head and Navigation Bar. It features different icons served for different purposes. For instance, under the Navigation Bar, it includes a Homepage Icon,

Defecting Reporting Form Icon as well as a Task View Icon. The head bar consists of the application name “Glidify”, Profile Icon, AI Chatbot icon, Help Information Icon and Notification Bell Icon. By having the fix interface panels, it provides a smooth connection throughout the application.

By naming the application ‘Glidify’, we hope that users get to experience a seamless glide through the work carried out in FM Operations. The Profile Icon feature was inspired by the modern apps in the current market. Envisioning to create a function to provide personalisation and keep close-customer service relationship with our users to enhance the overall user experience, the Profile Page becomes the perfect feature for us. (refer to 3.3.3 Profile Page for more information).

The AI Chatbot icon is another feature inspired by the current advancement of Artificial Intelligence (AI). Our team have decided to make use of its skill to upgrade the user experience throughout the application. (Refer to 3.3.6 AI Chatbot for more information).

The Help Information Icon serves as a quick instruction page more commonly used for first-time users. It is one of the features we thought would be perfect for beginners as it provides a quick rundown of the features we want to highlight and make sure users understands how to use them. (Refer to 3.3.4 Help Information for more information).

The notification Bell icon consists of two different types of notification feature, in-application and push notification. The purpose of having both is to allow versatility and ensure all users are invariably notified on new updates. (Refer to 3.3.2 Notification for more information).

In terms of the designing aspects, our team did some research on the various type of interfaces there are in the market right now. From fashion online websites to in-app services, we have attempted several different designs which have been improved consistently to provide only the best version we could offer. After multiple attempts, we have decided to go with the minimalistic approach to avoid clashes with the numerous data presented. All icons are aligned and sized to fit the overall look of the ‘clean girl’ aesthetic we envisioned.



Figure 30: interface panel (Header and Navigation Bar)

3.3.2 Notifications

a) Push-notification

When a form is submitted, a notification will be sent to all users from PowerApps. Having the Push notification sent directly to the lock screen of the user's mobile phone, reduces the need to enter the app continuously to receive notification. The message content includes the defect information, such as the defect issue, location of defect, and other relevant information that allows users to see its full details at one look. This would be good for users as they are still constantly updated on what is going on even without needing to physically enter the application to check.

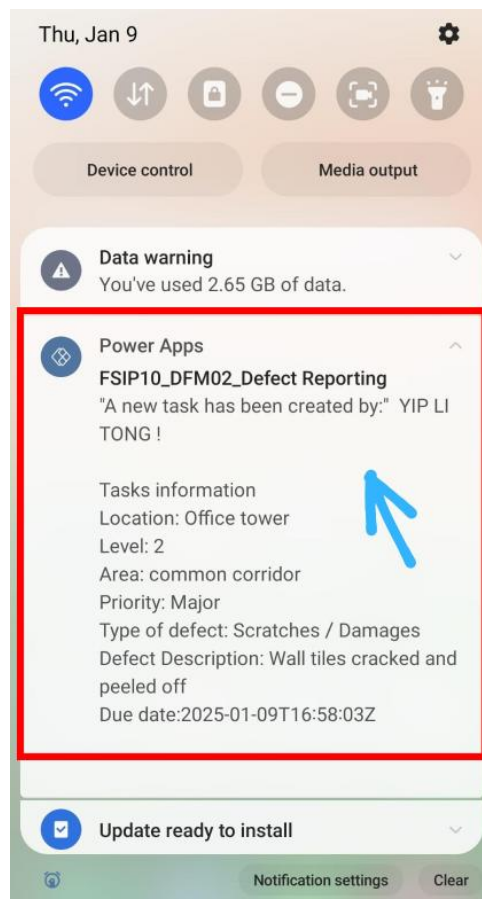
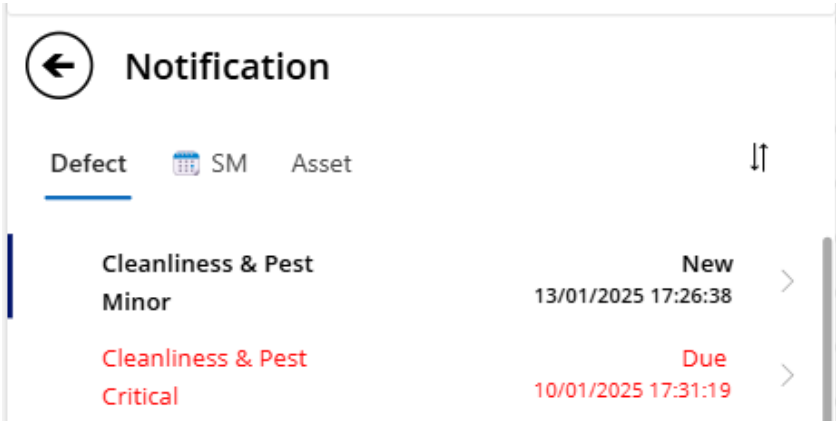


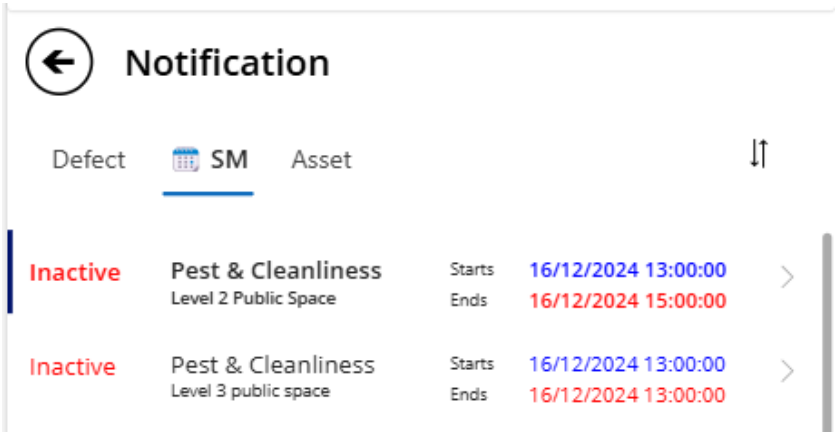
Figure 31: Push-notification from phone's screen

b) In-app notification

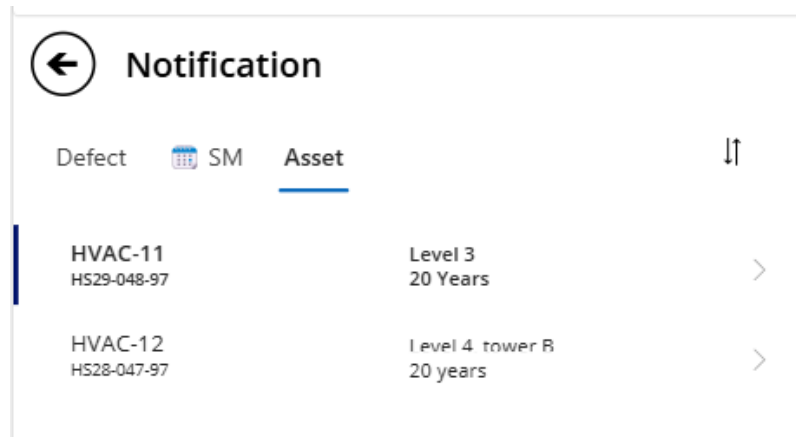
An In-app notification would be for when users are using the application and needs to check on the specific notification. Upon tapping on the notification icon at the top right corner of the app, a pop-up notification window will show up in the middle of the page. The notification consists of tabs, where users can conveniently switch across from one type of reminder to the other. A sorting button on the top right corner helps to sort the date from the latest at the top of the gallery, vice versa. By clicking on a specific notification, this would bring them directly to the information they need without too much hand movements.



Defect Reporting Form's Notification



Schedule Maintenance Form's Notification



Asset & Inventory Notification

Figure 32: Notifications User Experience

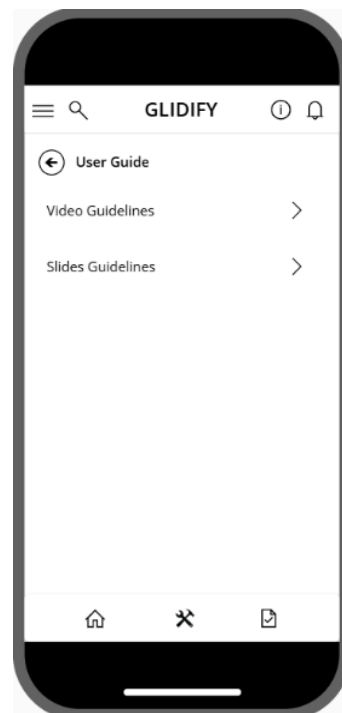
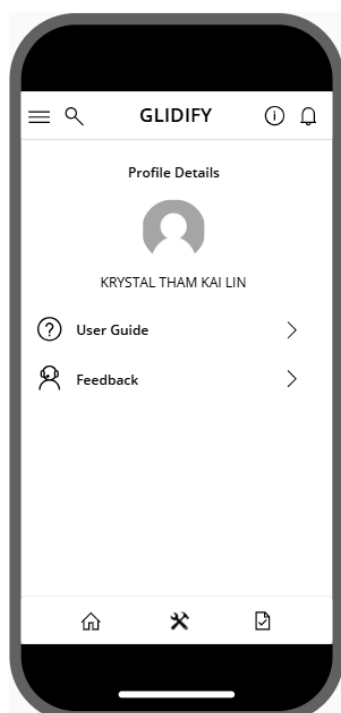
3.3.3 Profile Page

The Profile feature is a space for users to customise and understand the app by managing their information and preferences. Taking into considerations that all apps on the market have this common feature, our team strongly voted to include this aspect as one of our features as it allows for personalisation for users. For instance, we have pushed for the user's profile picture and name to appear on the Profile Page according to their emails which creates more personality and brings more familiarity.

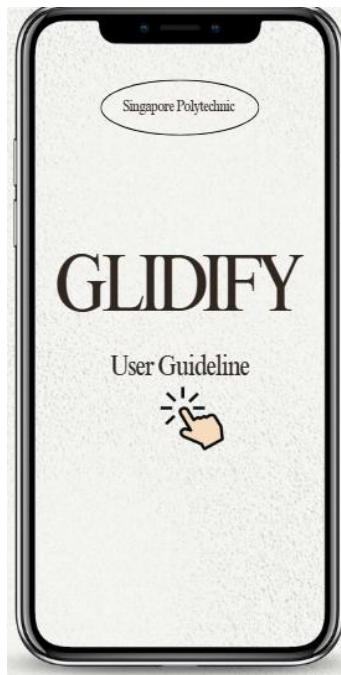
Apart from that, we have included two main features which we believe is most beneficial for our users. The first would be a User Guide which includes the Slide and Video version of the user guideline. Considering the audience we are targeting; FM industry generally consist of employees of a wide range of ages. It is therefore important to ensure that all users are provided with an instruction slides to help them out whenever necessary. It is also important to take note that this feature is different from the Help Information button as it provides an in-depth explanation and steps to fill up and submit every form. The

Slide version allows them to easily scroll through to find a specific page that they require while the video version is available for those who are illiterate or have difficulty reading.

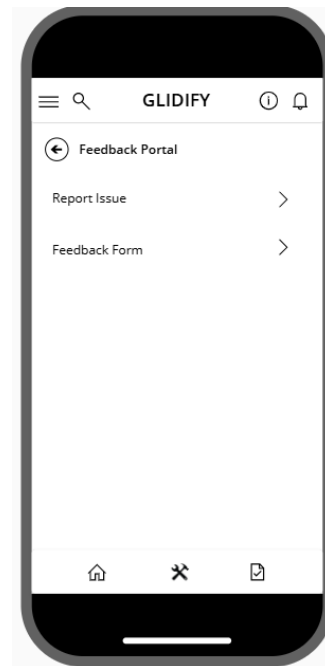
Next, we have the Feedback Portal which direct users to two different services. One of which is the Report Issue form whereby users can fill in any issues they face while using the app. This is more of a quick and go service for users, on the other end, our team will focus on working on improving the application in real-time without any downtime. Another feedback form which is the very last task users are encouraged to fill in for the company trial. Instead of providing a link to the company, we thought that it would be unique and easier for them to access the same application to fill in the feedback. It includes a list of questions that have been carefully crafted and designed for the app itself. With this form, our team will consolidate the feedback and provide a summarised version of improvements or additional enhancement which will be reflected as one of our future improvement works.



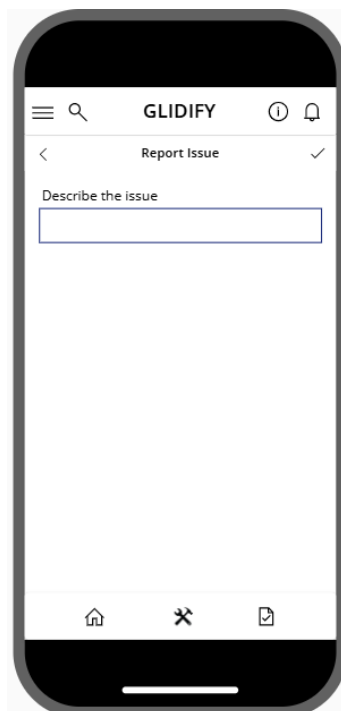
Profile Main Page



User Guide

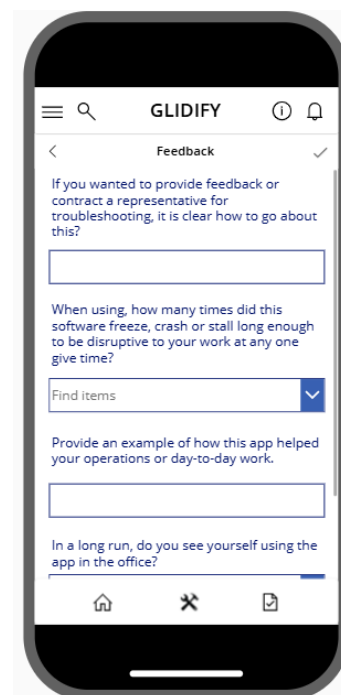


User guideline (Video/Slides)



Report Issue

Feedback Portal



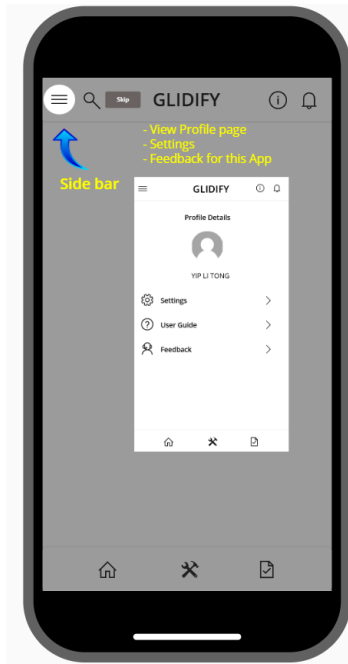
Feedback Form

Figure 33: Profile Page

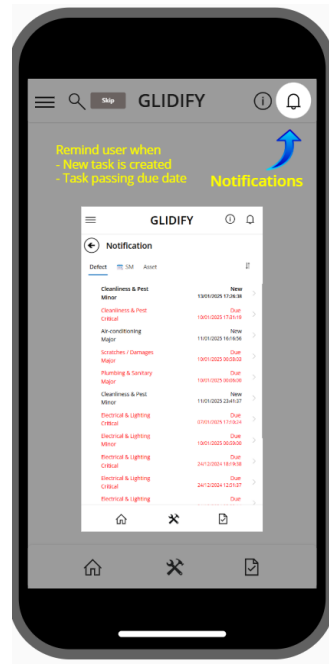
3.3.4 Help Information

The idea of including a Help Information feature came about after deep considerations of the target audience and age group we are focusing on. Many people in this industry might be less of a tech-savvy, thus ensuring that they receive a step-by-step guide on how to use the basic required features such as the defect reporting form, defect information and updating of the completed defect work will be beneficial to their day-to-day tasks. Above all, for users who are quick to learn, we have a function for them to ‘skip’ through the instruction page if they do not want to receive the full experience.

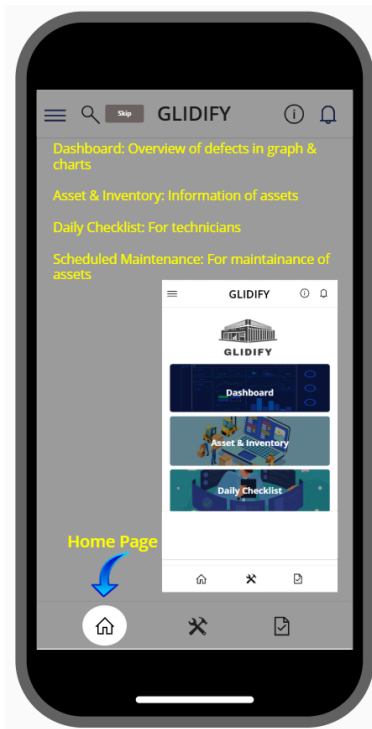
In any case if users would like to revisit the instructions or first-time users, they can tap on the top right-hand corner to review the feature. A quick instruction page will be reflected on its own highlighting features that are commonly used by our target group. All instructions come with photo attachments and explanations which we feel would ease the overwhelmed experience for older users as they are less knowledgeable on technology. This is an alternative to the User Guide.



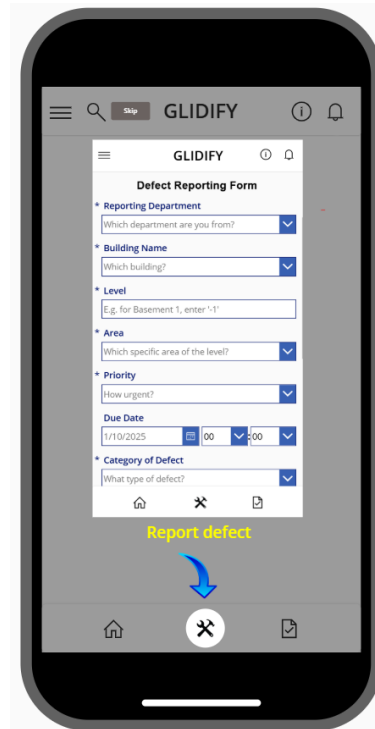
Page 1



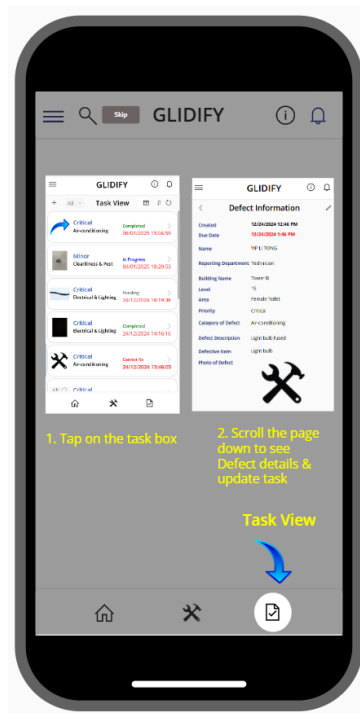
Page 2



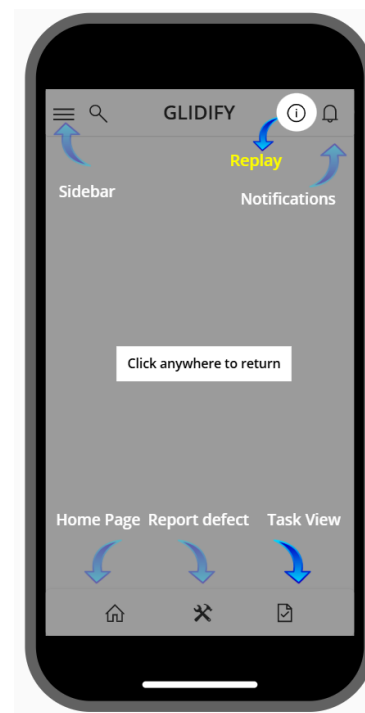
Page 3



Page 4



Page 5



Page 6

Figure 34: Help Information

3.3.5 Home Page

The Home Page appears just by a click of the house icon at the bottom left of the interface panel. The Home Page will be used as a navigator to bring user to portal of their choices.

It contains four major elements:

- 1) Dashboard
- 2) Asset & Inventory
- 3) Daily Checklist
- 4) Schedule Maintenance

In terms of functionality, the user interface is simple. For instance, by selecting the Dashboard, it navigates the user experience to the Dashboard Main Page.

One of our unique features of the home page is that, when tapping on all elements excluding Dashboard, it brings the user experience to another application. This portion

of the experience was inspired by SP's ABE Lecturer's app where they would be redirected to a separate interface. The advantage of this redirecting would lower the chances of the application hanging due to overloading of the data connected to our database. In case if the company wants to restrict certain users to certain elements, it can also be done easily without the need of major changes to the overall system.

On top of that, when it comes to the designing of the homepage, our team have taken inspiration from the different applications in the market. Focusing on neat and tidiness, we have chosen to include a generic picture of the type and name of element we are presenting. Keeping the aesthetic to minimal, it brings out the overall harmony we envisioned.

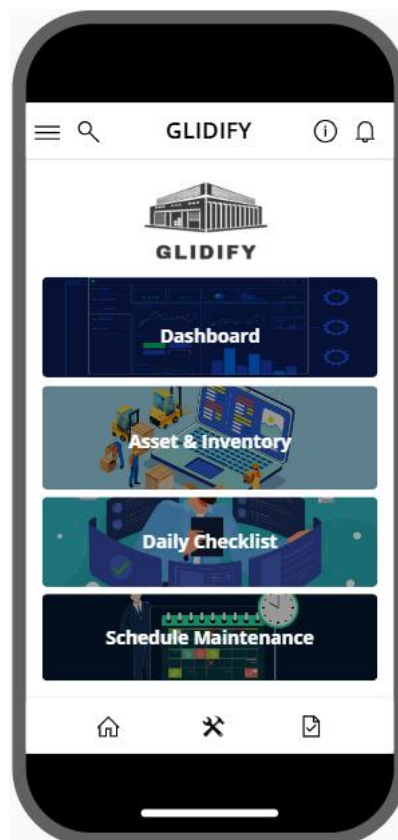


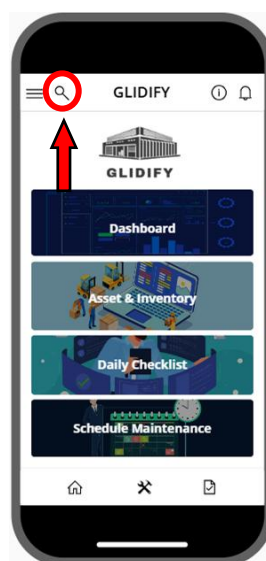
Figure 35: Home Page

3.3.6 AI Chatbot (FAQ)

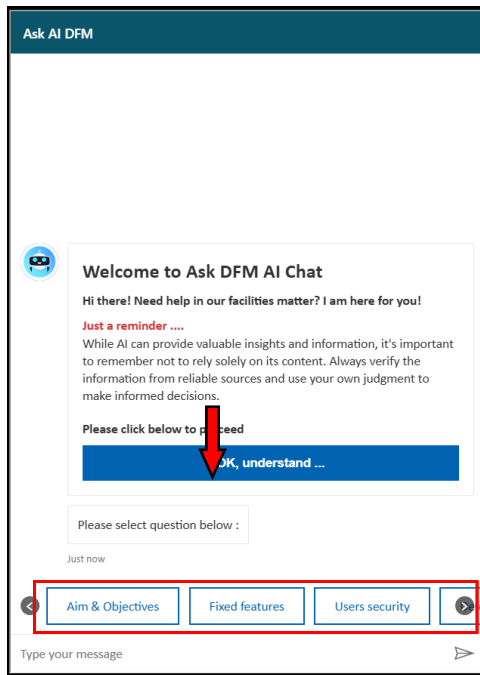
The Artificial Intelligence (AI) Chatbot was an enhancement feature added on to further push ourselves with the development of the application. Considering how advanced and common AI is used now, our team decided to make use of its skill of quick data gathering to enhance our user experience.

Similar to the workability of the Microsoft Copilot, just by a tap on the ‘search’ icon at the top left hand of the Header Bar, user will be re-directed to the AI chatbot to communicate with our AI personnel. User is allowed to ask questions related to our app for instance the security, aim and objectives, features that we provide. Question responses will be based on the script provided.

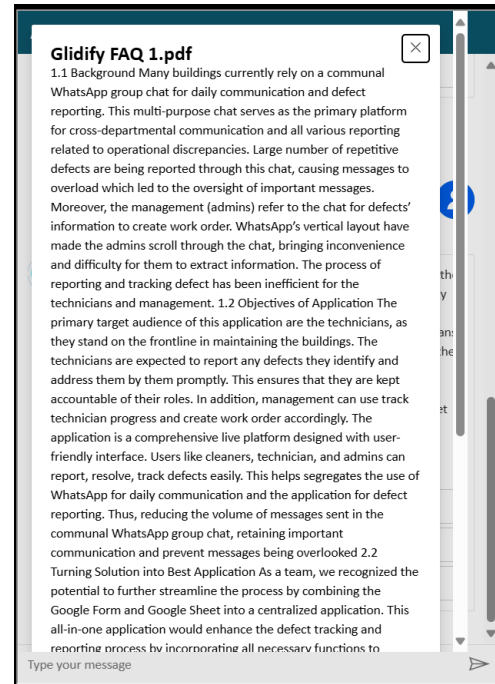
This feature comes in handy whenever technicians or other users requires specific information quickly. It would be useful for them to prompt the AI chatbot for hints or even answers related to the application itself whenever they have questions.



‘Search’ Icon



AI Chatbot



Reference from Script

Figure 36: AI Chatbot

3.3.7 Database

The Database stored in SharePoint contains all our required information created through Power App. All data are collected and updated in real-life, enabling management to keep a record history of all changes made throughout the accessibility of the application. The SharePoint database also reflects specific items such as the exact date reported, the exact date completed and provide modification dates which are interconnected to the information from Power App.

SharePoint

FSIP10_DFM02_Defect Reporting_From_M365

Private group Not following 8 members

Home Conversations Documents Shared with us Notebook Pages Feedback Form Schedule Maintenance... Site contents Report Issue Form Defect Reporting Form Asset Management HV... Asset Management Pr... Asset Management Of... Asset Management Ot... Inventory Managemen... Inventory Managemen... Return to classic SharePoint

+ Add new item Edit in grid view Undo Share Export Forms Automate Integrate

Test

No photos of defect + Add view

ID	Name	Report...	Building	Level	Area	Priority	Category	Defective...	Defect De...	Photo of ...	Working ...	Phot
1	Null 1	Cleaner	Tower A	16	Toilet (F)	Minor	Cleanliness & P...	Minor	Mirror is dirty		Completed	
2	Null 2	Technician	Tower B	7	MBE service room	Minor	Electrical & Ligh...	Light bulb	light bulb fused		Pending	
3	Null 3	Security	Tower A	20	Common corridor	Critical	Cleanliness & P...	shelves	Cockroach found on shelves		Pending	
4	Null 4	Security	Tower B	18	Toilet (M)	Major	Plumbing & San...	tap water faucet	3rd sink from male toilet entrance, tap water is leaking		Pending	
5	YIP LI TONG	Technician	Tower A	22	Toilet (M)	Critical	Electrical & Ligh...	Light bulb	Light fused		Pending	

Figure 37: Database

Chapter 4 – Problem & Skills Gap

4.1 Skills Gap

With no prior experience with coding or programming, developing an application from scratch is a tedious and challenging process. To bridge this gap, we relied on YouTube tutorials and search engines to learn and understand the necessary formulas. However, since no exact formulas perfectly suited our application's needs, we analysed available resources and adapted them to fit the app's features. While searching online provided useful guidance, its solutions were often inaccurate or too generic, requiring us to experiment and refine the formulas independently.

4.2 Unavailability of extended services

We encountered issues with licensing requirements for tools like the Power BI Dashboard and Copilot Studio Agent. We initially aimed to create a dashboard using Power BI because it provides a wide array of filter functions and customizable elements. We also intended to integrate an AI chatbot model using Copilot Studio Agent to analyse defects and provide summarized reports in natural language, allowing users to interact with the AI directly. Unfortunately, administrative restrictions on licensing prevented us from accessing these tools. Despite this limitation, we managed to find alternative methods to implement some of these features by combining elements within Power Apps and applying formulas to bring these features to life.

The alternative methods we have come up with for Dashboard feature was to create the graphs and charts from scratch. By making use of the in-app charts provided by Power App, we researched and found videos which guide us to create codes that suit the type of information we want to present on the experience. Although this method managed to present

the items we want, there were still limitations to how much we could show. One of the major falls back was the limited types of graphs and charts provided. In Power App, there was only three basics: Pie, Line and Bar graphs. Due to the lack of variety, we got to compromise with our initial plan of using stack charts and graphs which we believe would better present the data we want to showcase.

4.3 Limitations within the application

4.3.1 Dropdown Selection Errors

The dropdown options in the form are located close together on the screen, increasing the risk of users accidentally selecting the wrong option. If the form is not thoroughly reviewed before submission, incorrect data may be stored in the system. Since submitted forms cannot be edited, the user must contact the administrator with backend access to the SharePoint list to correct the information.

4.3.2 Issues with Images and Videos

Including images of defects in the form helps technicians identify issues more effectively and allows management to verify completed work. However, making the 'Image' field mandatory is not feasible due to the following limitations:

1. Power Automate Flow, which sends push notifications when forms are submitted, cannot capture images.
2. Uploaded images often have very low resolution, making them blurry and difficult to interpret.

3. Not all defects are visually observable, such as slow water leaks from taps. Capturing such defects is impractical since the leaks are intermittent and difficult to photograph at the right moment.
4. Only one photo can be included.

4.3.3 Access and Security Issues

Both technicians and cleaners have access to the task information page, creating potential security concerns. Cleaners may inadvertently or intentionally update tasks meant for technicians, and vice versa. For example, a task involving electrical wires detached from the ceiling could be updated by a cleaner without proper authorisation, leading to misinformation or delays in addressing critical issues.

4.3.4 Managing Employee Access

Proper access control measures are essential to ensure tasks are managed by authorized personnel only. Without adequate measures, employees might abuse the application, such as updating tasks on someone else's behalf without their acknowledgement. For example, Person A might assist Person B by updating a task but fail to notify Person B. If management reviews the task and finds it unresolved, they will hold Person B accountable instead of Person A.

To prevent such conflicts, we created a label at the bottom of the task that records and display the name of the person who modified the form and the specific time when the form was modified. This helps management to identify the person in charge and deter malicious or unauthorised actions.

4.3.5 User Adoption Challenges

Older technicians and cleaners may face challenges in using the app due to a lack of proficiency with digital tools, making it difficult for them to learn a completely new skill. Some employees may also resist adopting the application and refuse to use it. While training sessions or one-on-one teaching could help familiarise them with the app, these solutions require significant time and effort, which is impractical given our tight schedule and limited timeframe.

Considering the constraints we face; we have come up with solutions which we hope could tackle this issue. We have created two user guideline versions, slides and video, which could help those who are not tech-savvy. The user guidelines are described simply with steps and visuals to showcase the specific areas needed to be filled up. Older users can refer to them whenever they need just by tapping onto the 'User Guideline' where they will be redirected to the selected version.

However, it is also important to hold a workshop with the older generation to guide them as first-time users. This lesson will serve as the first introduction to the application, aiming to share and teach them this new service at a pace that would not be too overwhelming to handle.

Chapter 5 – Feedback & Review (company)

Our team presented the demonstration application to our industrial link company and received a couple of recommendations from them. Based on their comments, we have summarised their suggestions below.

5.1 Feedback from Company

For a start, the defect reporting form is created by the user who could be of any member of the organisation. The person completing the task can be of another person of any member of the organisation. The company have suggested to include a section for the management of approve the work after the work is completed by the person fixing the problem. By creating an approval section, a push notification through email as a reminder. This creates a cycle of completing and checking, ensuring that the management is notified and acknowledges the job is done according to industry standards, preventing any unwanted confusion among colleagues.

Secondly, they would prefer to have a summary report of the Daily Checklist reflected or outsourced onto a website for ease of referencing. In simple words, it would be vital to review the report in a format similar to excel but with a twist of having the listed details on the left and the dates created on the top row. Our team have explained the limitations associated with the materials we are using. Thus, there was no changes made to the final version.

Thirdly, they have suggested to include the ‘modification’ feature in the Asset & Inventory page where users can see when someone adds or remove an inventory. In a way, this would provide clarity on the last user who used the items in the inventory and prevents unwanted misunderstanding of blaming the wrong person when items are misplaced.

Suggestions recommended by the company will be considered and improved consistently to provide the best possible version we could offer.

5.4 Improvements

Following the suggestions provided by the company, our team have developed an improved version of the application to customise to the requirements.

Improvement 1: Approval Section

For the first problem, we have included both the approval and email function to tackle the issue of ensuring that the work done is kept at a certain industrial standard checked by the management. Upon completion of the work, the user fixing the issue will have to select 'Approved by' from the drop-down menu and select 'Send Reminder' to their direct supervisor or management for approval on the work done. The dropdown selection choices for the 'Approval' field includes 'Approved', 'Not Approved' and 'Pending for Approval'. Below the choices is a textbox where users can leave comments or remarks about the quality of work completed. Next, the name of approver is automatically filled in the 'Approved by' field. Finally, users can customise the message and click on 'Send Reminder' button on the top right corner of the rich text input, a popup box will show the names of all members using the app. Clicking on the specific user will trigger to send the customised email message to the selected person. The selected personnel shall conduct the quality check to ensure it meets the necessary requirements before this report is fully closed. This ensures accountability and maintains quality standard of work delivered.

Improvement 2: Sending email to specific user

To address the problem of communication between the technician or cleaner and the management, a text input was added into the approval section, allowing two-way communication whereby the user to send email to specific person and receive emails sent by other users. For instances, once the form is submitted; the supervisor shall receive an email notification notifying them on the updated work. In case if there is no need for an approval, the user fixing the issue can choose the usual route of submitting the defect with photo attachments marking the work 'Completed' and click the tick button at the top right-hand corner to update the form.

By having this two-system updating function, it gives user a choice to either submit with or without approval from their management. As such, it prevents any excuses allowed as there are several methods utilised to complete the form. It will fully help us to achieve the objective of ensuring members of organisation takes accountability of their own work and ensures the quality is met.

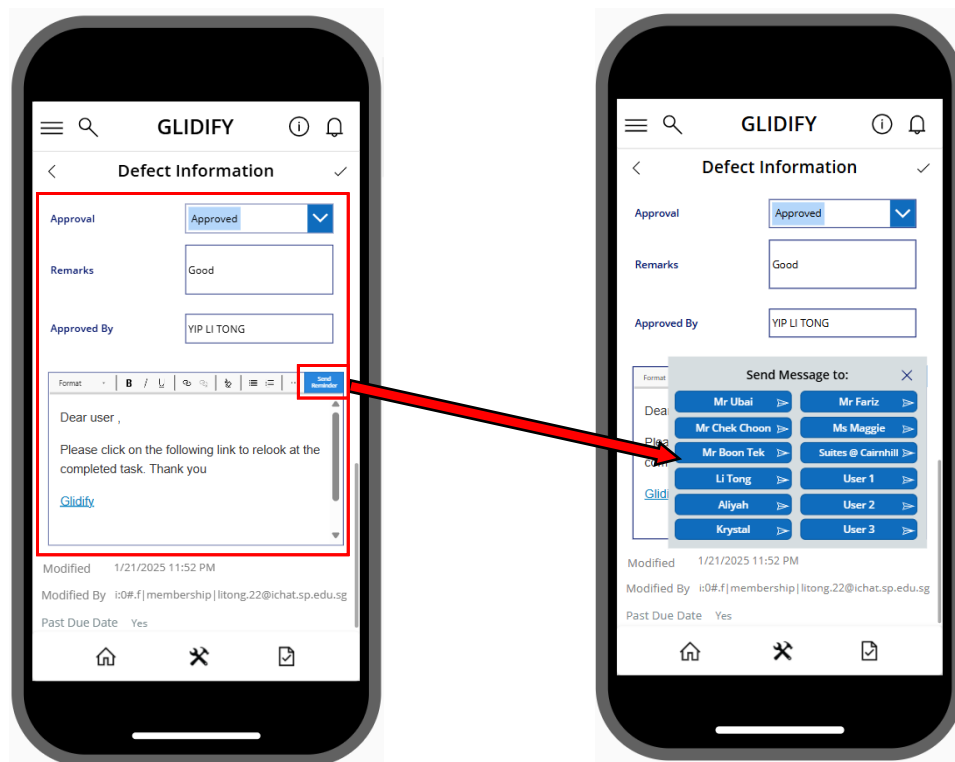


Figure 38: Task Information (Improved version)

Next, for the modification update function in the Asset & Inventory, our team have identified an automatic recording of the modified version in our database, SharePoint where it shows the modification date and time as well as the last modified personnel. The issue would be how we work on to interconnect this functionality onto the application platform.

As such, we have come up with a method to include the modification settings into the user experienced. When the user opens the Inventory details, there will be a modification date and last change personnel reflected on the screen. The update will be aligned to the database continuously.

Chapter 6 – Conclusion

6.1 Reflection (Individual or Team)

6.1.1 Team's reflection

Throughout this project we have gained new insights on how the adoption of technology gradually overcome the traditional facilities management. This project encouraged us to go extra mile, understanding the how the tools and resources are customized to support streamlining processes in facilities management operations.

Besides, we also learned that the practicality of bringing the ideation into reality may sometimes differ from our expectations. Initially, we planned many complex features associated with application. However, we overestimated our time and ability to produce the planned work, which caused delays in delivering our work. This led us to review our initial plans and make further adjustments.

As we step into the future, we hope to make a difference in facilities management industries through adopting technology, such as our idea of integration of Artificial Intelligence into the application to stay relevant in the age of technology. We hope to bring the best version of our app to help make a positive change in the facilities management operations.

6.1.2 Individual reflection

Li Tong:

After exposing to the application's backend configurations, I realised the importance of coding experiences and programming skills. Without the basic foundations, I struggled to understand the logical flow and the necessary codes for the application to work. I

could also relate to the hardship and effort input by other similar application developers. With perseverance and belief, I searched for different ways to overcome problems and effectively utilised AI platform to explain the codes. This approach has greatly helped me to create an application from scratch.

I felt grateful for having an opportunity to lead the group as a leader. I have learnt to manage the team and work with different people, ensuring that the group is moving on the right track and progressing towards the shared goals of developing the application.

Finally, in group dynamics, I learnt that teamwork and coordination is very important to achieve consistency and ensure that all aspects of the apps are linked to one another. Clear communication and transparency between teammates are equally important to keep everyone informed and updated.

Krystal:

Overall, the hard work and effort put into the Integrated Project felt like a bittersweet moment. I joined the group with lots of confidence as we all aimed for the same goal which is to receive an exceptionally well grade. Knowing my limited capabilities in the develop and programming of the app, my team members were understanding and compromising. Still, I tried my very best to take charge in other aspects such as the GUI, admin work, researching and helping with simple programming to my best ability. Nevertheless, this project has taught me that programming was not as scary as it seemed, it is more about understanding the long and foreign logics to achieve the information I want. Taking the lead to design the interface was lots of fun, it was something different from the usual module tutorials which really spark my creativity and designing mind.

I also really enjoyed planning and researching for the app. It was a module that I did not feel exhausted even after working into the night for it. I truly enjoyed creating an app from nothing, but unforeseeable circumstances arise which caused burn outs and mental exhaustion. I honestly did not expect a burn out to happen considering how enthusiastic I was. But accepting that fact, it took days to get back on track which caused me to lose time which should have used to develop and improved the app. On one hand, the break was necessary, but on the other hand, it added more workload to my team member who already had a lot on their plates.

Another thing is that, from discussion to planning to teamwork, I learnt a lot about myself and how I am as a team member. As someone who frequently works with groupmates of variety of personalities, I grew to become one who separates work and personal emotions well.

However, I realised how affected I was throughout the project execution and could not put my emotions aside at some moments. There were lots of self-doubting moments that changed my perspective of my role as a team member. I begin to recognise what I seek for as a teammate and from my team members. Communication is my top priority in projects that requires such high demand in discussion and planning. Having to experience a total opposite from what I have expected, the miscommunication draws several emotions that resulted in me feeling scared to even make amendments as I was afraid of being constantly framed for 'mistakes' I did not do. It caused me to constantly question my own actions and words, thinking that I was the biggest 'problem'. When it finally

occurred to me that there was no victim here, the misunderstanding grew into burning hatred that couldn't seem to die down.

As a team member, I do think that certain approaches I have chosen were wrongly executed. However, it was the only method that I knew could temporarily fix the enduring issues to make sure the necessary work is still completed. I tried my very best to keep the professionalism and speak up about issues in hope to receive some active improvements. Yet, it was once again disregarded and ignored. The constant cycle of providing yet not receiving the same effort tires me and there is nothing more I can do to than to accept the fact and work on improving myself to prevent history from repeating.

At the end of day, I gained some and loss some. The process from start to finish truly tested my patience and endurance but it has also taught me new skills. Carrying on these experiences, I hope to become a better person and just stick to the famous saying of 'never working with friends' because it truly scared my entire soul away.

Aliyah

This project has given me the opportunity to think critically when creating something. The amount of research required and understanding the objective and target audience is very crucial to avoid side tracking. Project planning is also crucial not only to stay on track but also ensure we are able to cope with life. This project has drained me physically and mentally as I struggled finding the balance to commit to this group project, other schoolwork and personal life. Having back-up plan if things we to went sideways is key.

This is regarding time and completing all the targeted features of the project. This project has taught me the difficulty working in a group.

Communication is of high importance to ensure that everyone is on the same page. The needs to keep each other up to date regularly even for the smallest changes is important as it may cause a chain reaction affecting the other areas of the project. Speaking up to justify your ideas while finding the balance to give in and accept other suggestions is not black and white all the time. At the same time, facing rejections is expected but the needs to listen and understand the reasoning is significant as one should not expect others to accept their work if it not up to their expectations or in line with the project works. Aside from them, informing other of one's struggle is necessary. As much as one would like to give their best, one should tell others if they need help instead of expecting them to constantly ask as this would eventually impact the whole team progress.

From this project, I realise the importance of helping one another when working as a team. As far as one should expect quality from other groupmates work, the need to be open to accept others' limitations is critical. It is important not only to help one when they are struggling but also when they make a mistake. Instead of disregarding them, acknowledge their mistake and help make it right as one has their own flaws too that they may or may not realise. This can help prevent breaking one's spirit, resulting them to second-guess their every action and be wary of making a mistake. Additionally, supporting and understanding other needs for time it is part of helping one another. One should respect other time when they are unavailable or need a break, within reasonableness, and help lighten their workload instead of expecting them to continue doing their part fully.

Lastly, I learn the hard way to listen to the saying “Don’t work with your friends” . This saying is especially applicable if one does not really know their friends. Finding the balance to support your friends while keeping professionalism to ensure the project success is not as easy as it seems. At times, the boundary of friendship and professionalism can get blurry but regardless, respect to one another should remain intact. Being patient with one another is crucial not only as friends but also as groupmates but there is a limit to it. Part of me is glad that this project has open my eyes up to ones’ true colours and the perception of who I am to them, a friend or a groupmate. This got me realising who I can truly call friends and who is just hiding behind a mask.

In the end, I believe as a team member, I have my own fair share of mistakes. Being more patient and receptive to others was one of my struggles at the start but I attempt to overcome over time. Taking action to fix disagreements instead of hoping it would eventually resolve is another error. Keeping my emotions in check is part of it too. Undeniably, I wished I could turn back time to make things right and prevent the unwanted chain reaction that resulted in a fall-out. Nevertheless, it will be a lesson I keep close to myself.

6.3 Future Prospect

From the suggestions of the industrial company and lecturers, we learned that we could add on more features in our app to enhance its usefulness and practicality. The following features are some of the areas we explored but were not done.

6.3.1 Safety Checklist

With the importance of Work, Safety and Health (WSH) regulations enforced during renovation, A&A works etc. A safety checklist is vital to be implemented as it allows building managers to cross-examine their contractors. This recommendation would ensure that contractors abide to the regulation when carrying out certain daily or scheduled duties which are often flagged as dangerous.

Additionally, it is also a method to ensure users take accountability of their own actions. In any case, if there was an accident, this safety checklist can be used as evidence to receive proper coverage. Overall, the safety checklist would be feasible in big companies where several works are being conducted in a day. At times when certain safety issues are being overlooked by the inspectors, they could use the checklist to prevent side tracking. This feature not just ensures users follow the regulations but also prevents them from oversight of points.

By having this would streamline the whole process as user would start off by daily checklist, followed by reporting defects, scheduling for new works to be completed before checking on the maintenance that are undergoing improvements. Every step requires a safety check to align closely to the rules to prevent one from getting into unwanted trouble or unsafe acts.

6.3.2 AI Chatbot (for analytics)

Integrating a virtual agent chatbot that can answer to questions related to the data within the app, helping managers to analyse the data and answering queries in seconds upon sending the question. It also provides a detailed description of defects

pattern as well as summarising the overall defect reported. This feature comes in handy whenever technicians or other users requires specific information quickly. It would be useful for them to prompt the AI chatbot for hints or even answers instead of going through the application to look for what is needed. Overall, it helps saves time spent and allow users to receive data immediately.

As a group, we have tried to integrate AI models into the application using the Microsoft Studio Agent. However, we have come to a dead endpoint where the access to it is restricted by the administrator. With resilience, we tried to invent and integrate an AI model using free API keys found from Github and requesting using the 'Post' method. However, we couldn't figure out the endpoint of the API key, which is crucial for identifying the digital location where an API receives requests about a specific resource on its server.

6.3.3 Dial Call Function

Adding a call function will enhance the communication between cleaners, technician and management team. At times, when a defect is reported, it could be difficult to describe it through text. Hence, having a call function allows enhancing clarity and provides a clear communication platform between the operations team. To enhance the experience, it can be interconnected directly to the phone's caller function which would easily allow user to make calls without needing to manual leave the app.

6.3.4 Offline Application

To enhance the usability and functionality of the app, our team have suggested to upgrade the application to an Offline application for future improvement work. It is common for Facilities Managers, Technicians or Cleaners to come across poor Wi-Fi or data connections, resulting in possible delay of work submissions. Thus, to simulate a seamless functioning app even under the limitations of poor server connections, making the application functional offline would be vital.

6.3.5 Language Feature

During internship, our team have worked with a variety of people of different races with languages limitations. Thus, one of our thoughts when working on this application is to accommodate to everyone's difference in language knowledge. Although, it was not implemented at this stage, we hope to bring this feature in as future enhancement. Having able to change the language settings of the app, would ease the user experience.

Similarly to Iphone's language function, there will be a minimum of 32 languages added for user to choose. By prioritising Singapore's main languages, Chinese, Malay, English (UK) and Tamil, it will present the personality that our team want to showcase in our application as 'to be seamlessly friendly and accessible'.

6.3.4 Duplicator Checker

One of the common issue our team have experienced during internship is the constant duplication of work. This arise whenever there are overlapping information, resulting in employees having to do twice the amount of work. To reduce the lightly hood of this

occurring in near future, we suggest implementing a duplicator checker which flags out duplicated copies of photo attachments.

A real-life example for visualisation purposes would be the Iphone Photo settings where they would prompt the user on the existence of two similar photos and put out a request if they would prefer merging the copies together.

Taking inspiration from this, it would be beneficial in a long run as management do not have to go through the long list of data to delete the duplicated copies. At the same time, this also makes the dashboard summary more accurate and reliable.

6.3.5 Enhancement on Filter

It would be great to include a filter function to sort the data base on the location of the building. In a long run, if this app gets implemented, it would be feasible to create data base on the building the user is in to prevent overlapping of information or confusion. This would aid in the clarification and strength the purpose of fully utilising and streaming the entire application that includes all possible features to increase the efficiency of the FM building.

Appendix/Annexes

Appendix 1: GA2 Programme Chart90

Appendix 2: Approval of Proposal from Company91

Appendix 3: Industrial Linkage Proposal & Slides92

Appendix 4: User Guideline93

Appendix 1: GA2 Programme Chart

Programme Chart for DFM Integrated Project

Names of Students:		Yip Li Tong			Krystal Tham Kai Lin			Nur Aliyah Umairaa Binte Riduan			NIL			Date of Submission:		
Project Title:		Streamlining Defect Process												5:00 pm, 24 Jan 2025		
Facilitator:		Mr Ubai														
S/No.	Description of Activities	Week														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	Research of product															
1	Obtain feedback from current users															
2	Research on possible app developer															
3	Research on related software in current market															
4	Research on users needs and functions															
5	Research on questionnaires for app development(what can be considered?)															
	Development of prototype															
	Phase 1															
6	Creation of Form															
7	Creation of Task View															
8	Creation of Login Page															
9	Creation of Profile Page															
	Phase 2															
10	Creation of Homepage															
11	Asset & Inventory Feature															
12	Creation of Daily Checklist															
13	Schedule Maintenance Feature															
	Phase 3 (Implemented when Trial)															
14	Safety Checklist															
	Phase 4 (TBC)															
15	Offline Feature															
16	Other Improvements															
	Product Testing															
17	Phase 1 Testing by Team Members															
18	Phase 2 Testing by Team Members															
19	Overall Testing of Phase 1 & 2 by Team Members															
20	Trial by Company															
21	Phase 3 Testing by Team Members															
22	Feedback from Company															
	Others															
23	Proposal to Company															
24	Finalise objective/scope of work															
25	Submission of GA1															
26	Finalise and submit GA2															
27	Submission of GA3															
28	Finalise and Submission of Draft Report															
29	Finalise and Submission of Final Report															
30	Update Report															
31	Update Journal															

Appendix 2: Approval of Proposal from Company

4. PowerPoint guide & instructions

- Briefing to use the form (Video to use the form & PowerPoint guidelines)
- Maintenance of Toilet Defect System (PowerPoint to maintain both form & database)

Trial Period: 1/8/2024 to 7/8/2024

During this trial period, the system will be implemented with continuous monitoring to ensure its operation aligns smoothly with the planned objectives. The form and database will be subject to ongoing revisions throughout this timeframe to enhance convenience and functionality.

Actual Implementation: 8/8/2024 onwards, the Toilet Defect Reporting system will be officially implemented, replacing the previous 3-in-1 WhatsApp group chat used for reporting. The system is not to be disclosed publicly, and the prototype cannot be brought back to school. Instead, a video will be provided solely for the purpose of demonstrating how the system works.¹²²

To verify the implementation of the system, kindly request your endorsement on this document. Your signature will serve as confirmation of the system's deployment. Should you need any additional information or have any questions, please do not hesitate to reach out. Your cooperation and assistance in this matter is greatly

appreciated. To confirm your acknowledgment and agreement, please provide your name and signature below:

Proposed by: Yip Li Tong (P2225546) Class: DFM/FT/3A/02
Date: 8/8/2024

Name of person in-charge: Swia

Signature & Date: Swia 8/8/24

Company Stamp:



Appendix 3: Industrial Linkage Proposal & Slides

Official [Closed], Non-Sensitive

Integrated Project (IP) Proposal

Our Team

Name	Admission No.	Email
Yip Li Tong (Leader)	P2225545	ltong.Y2@schol.ap.edu.sg
Nur Aliyah hugaga	P2225757	alyahnurasa.Y2@schol.ap.edu.sg
Krystal Tham	P2225559	krystaltham.Y2@schol.ap.edu.sg

We're writing this proposal in hopes that you will support our integrated project (IP) for our final semester. The IP aims to develop student's initiative, self-reliance, creativity, innovation, entrepreneurship and teamwork by having them work independently in a simulated work situation.

As a valued Industry Partner, we hope to collaborate to conduct a trial and testing of our prototype/proof-of-concept (POC). Your valuable feedback will help us improve the prototype in the hopes to help improve the efficiency of facilities operations.

Our IP is a prototype of an application which aims to streamline defect reporting process and other features such as Daily FM Operations and Maintenance. This app is made for technicians and cleaners to take accountability of their duties and roles.

Detailed Breakdown of the Software Features

- Defect Reporting**
This feature consists of a set of questions that help generate Work Order automatically. It includes necessary details like locations, urgency, photo of defect and other possible related questions. Thus, this would minimise the need of Administration to create Work Order. To track the defect reporting cases, the responses will be reflected in a database. This database will be accessible to the management.
- Task View**
Any work order generated or required maintenance would be listed in the Task View for required personnel to complete. There will be Filter feature in the Task View to organise the work according to their urgency.
- Push Notifications**
According to the urgency of the defect rectification, there will be notification alerted to all users of the app. If not done so, the app will send reminders through push notification to technician as well as supervisor in-charge for monitoring. This will be included in features our Asset & Inventory and Scheduled Maintenance.

Possible Additional Features

- Asset & Inventory
- Daily Checklist
- Schedule Maintenance

Official [Closed], Non-Sensitive

Breakdown of Trial Schedule:
The duration of the trial will be about 4 to 6 weeks.

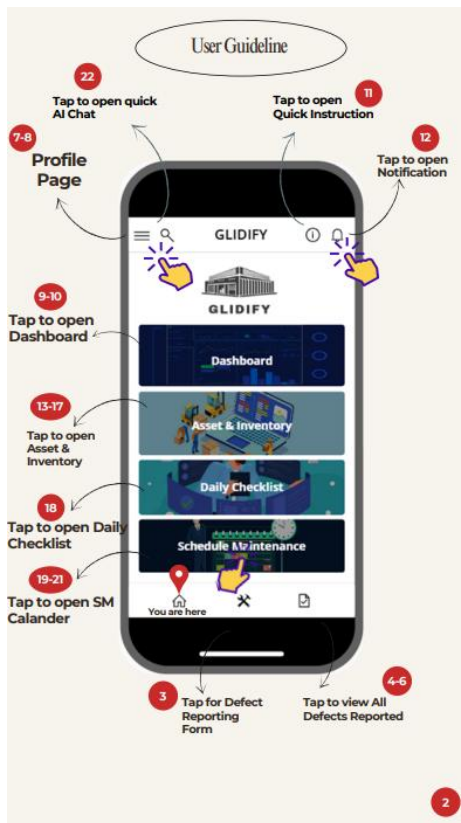
During this testing period, the software will be implemented with continuous monitoring to ensure it operates smoothly with the planned objectives. The app is subjected to ongoing revisions throughout the timeframe and feedback will be collected to enhance convenience and functionality.

Industrial Linkage Proposal

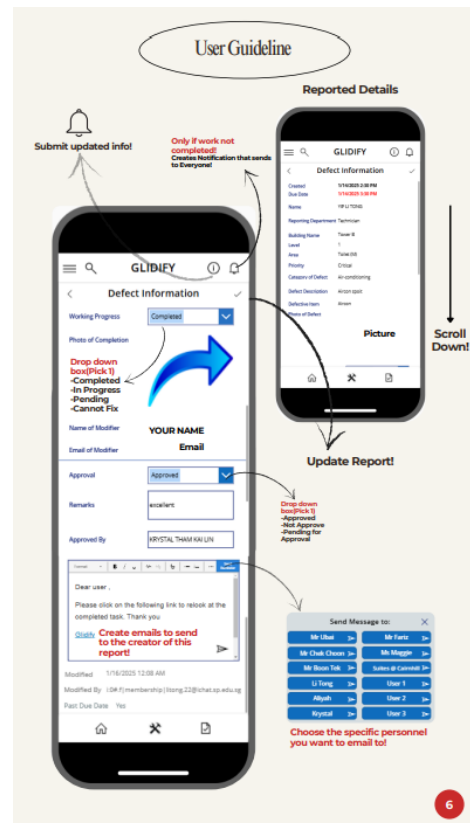
The presentation slides are arranged in a 2x4 grid. The first four slides contain content: Slide 1 shows a person reporting a defect; Slide 2 introduces the app and its goals; Slide 3 shows the defect reporting form; Slide 4 shows the product infographic with various app features. The next four slides are empty: Slide 5 shows a detailed task view; Slide 6 shows the app's features; Slide 7 and Slide 8 are blank.

Industrial Linkage Presentation Slides

Appendix 4: User Guideline



Content page with numerical page numbers
according to the pdf numbering



Reported Defect details & updating for approval/emailing from management



Deep explanation of the different icons we provide in the task view user experience



Step by step explanation on how to go about trying out our dashboard



Step by step to explain the functions and how to use our Schedule maintenance calendar to **review the details**



Step by step to explain the functions and how to use our Schedule maintenance calendar to **create the required work**

References (or Bibliography)

References for learning to develop application

1. Power, in (2024). CHART Controls in Power Apps | Line, Column, and Pie Charts. Available at: https://youtu.be/RfiXzEb1jwU?si=yrlRGhN8aTHpJ_25 [Accessed 7 November 2024].
2. Power, in (2022). Create Multi select Checkboxes in Power Apps for SharePoint List Forms/Radio, List Box control. Available at: <https://youtu.be/Vj8uP2zaMtA?si=vfggnT5T1AU2LeIH> [Accessed 23 October 2024]
3. Power (2023). Power Apps Tutorial-Responsive Screen with Gallery & Filters – Beginner to Advanced. Available at: <https://youtu.be/ubnPqwWdBL0?si=st5JYljFef1pGtcW> [Accessed 23 October 2024]
4. Push, C. (2021). Create push Notifications for Power Apps mobile – Tutorial. Available at: <https://youtu.be/YLYsognPA6s?si=NPO8kvjT9SdTMLgn> [Accessed 23 October 2024]
5. Power,B (2023). Building Power Apps Navigation & Flyout Menus with Modern Tab List Control. Available at: <https://www.youtube.com/watch?v=ZWl2jwC9p3w&t=364s> [Accessed on 7 January 2025]
6. Power.I (2022). How to make a login screen in PowerApps. Available at: <https://youtu.be/BN74SsN4mm8?si=1tsrggax-FD2Dag> [Accessed 5 November 2024]
7. Power, in (2022). Build CALENDAR Control in Power Apps in Minutes/ Connect to SharePoint List. Available at: <https://youtu.be/pA7pcAOvf10?si=kw0E0cmxqIJ0hTlx> [Accessed on 7 November 2024]
8. Power (2023). Power Apps Screen Designs (UI/UX) – Power Apps Tutorial. Available at: https://youtu.be/YPN6A_8N1Vk?si=9jBdxWG19qPb5Qe7 [Accessed on 10 December 2024]
9. Anil. (2022). PowerApps Data Entry Form | Enter Data into SharePoint List using PowerApps Form. [online] Available at: <https://youtu.be/y6ZorSP7C3Y?si=NvWkcCNQNnWObmOI> [Accessed 9 November 2024]
10. Reza D. (2023). How to apply Multiple Filters to Power Apps Gallery: Dates, Ranges & More!.. Available at: <https://youtu.be/1QflzfPyPe0?si=IDu4vHbLv4WHgfbid> [Accessed 9 November 2024]
11. Reza D. (2022). Power Apps Gallery Edit Form Tutorial for Beginners. [online] Available at: <https://www.youtube.com/watch?v=HHXKfB1iAH4>
12. Talk, L. (2023). Let's talk about the Combo box Again with PowerApps (and Modern!). Available at: <https://www.youtube.com/watch?v=ONPI9cf2JLQ&t=86s> [7 January 2025]
13. Power, in (2021). How to create tabbed forms in PowerApps. Available at: https://www.youtube.com/watch?v=KjUJZr5Vn_Y [6 January 2025]

14. Power (2023). Power Apps Editable Grid Using Gallery & Modern controls. Available at: <https://www.youtube.com/watch?v=76RJyaoW0BQ&t=136s> [2 January 2025]

References for Images used in application

1. Power app's Reporting defect icon:
Ahkam. (n.d.). Icon Vector Tool PNG Transparent Background – Freeiconspng. [online] Available at: <https://www.freeiconspng.com/img/8066> [Accessed 23 October 2024]
2. Help Information page guiding arrows:
Clipart. L. (n.d.). 3D Arrow Png #925092 [online] Available at: https://clipart-library.com/clip-art/5-57314_blue-arrows-png-arrow-in-png.htm [Accessed 20 October 2024]
3. Flaticon. <https://www.facebook.com/flaticon> (2021). *Month Icon* - 5776621. Available at: https://www.flaticon.com/free-icon/month_5776621 [Accessed 12 December 2024]
4. Stelian Subotin (2017). *Dashboard Design: Considerations and Best Practices*. Toptal Design Blog. Available at: <https://www.toptal.com/designers/data-visualization/dashboard-design-best-practices> [Accessed 12 December 2024].
5. LinkedIn.com. (2025). *LinkedIn*. Available at: <https://www.linkedin.com/pulse/understanding-importance-asset-inventory-saheed-oyedele/> [Accessed 12 December 2024].
6. Green, D. (2023). *Building Smart Preventive Maintenance Schedules*. Click Maint CMMS. Available at: <https://www.clickmaint.com/blog/preventive-maintenance-schedule> [Accessed 12 December 2024].
7. Bubble. (2025). *The most powerful no-code platform | Bubble*. Available at: <https://bubble.io/> [Accessed 10 October 2024].
8. Flutterflow.io. (2025). *FlutterFlow Desktop Apps*. Available at: <https://www.flutterflow.io/desktop> [Accessed 10 October 2024].
9. Figma. (2025). *Free Design Tool for Websites, Product Design & More | Figma*. Available at: <https://www.figma.com/design/> [Accessed 10 October 2024].
10. Zapier.com. (2025). *Automate without limits | Zapier*. Available at: <https://zapier.com/> [Accessed 10 October 2025].
11. Formassembly.com. (2023). *The Form Builder with Unlimited Web Forms and Submissions*. Available at: https://go.formassembly.com/unlimited-form-builderdemo.html?utm_source=google&utm_medium=cpc&utm_device=c&utm_campaign=Search++Competitor++All&utm_adgroup=137793518314&utm_extension=&utm_term=jotform&utm_landingpage=https://go.formassembly.com/unlimited-form-builderdemo.html&bt=589028512762&bk=jotform&bm=p&bn=g&bg=137793518314&gad_source=1&gclid=Cj0KCQiA7se8BhCAARIsAKnF3rwezrA1AmTQe_Mn

[22Oyvbvm0Ht6Ts5VSac1635HrlBRiTLGui04BB4aArd6EALw_wcB](https://www.googleapis.com/auth/cloud-platform) [Accessed 10 October 2024].

12. AppSheet. (2025). *Google AppSheet | Build apps with no code* Available at: <https://about.appsheet.com/home/> [Accessed 10 October 2025].